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MASTER

Machine learning based predictive maintenance and regression based control of a servo motor for industrial conveyor belts

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MasterThesis

MachineLearningbasedPredictiveMaintenance andRegressionbasedControlofaServoMotor forIndustrialConveyorBelts

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Iherebydeclarethatthethesissubmittedismyown,unaidedwork,completed without any unpermitted external help.Only the sources and resources listed were used.

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Abstract

Industry4.0revolutionandthetechnicaladvancesinInternetofThingsresearchbrings us smarter and more connected sensors and devices for industrial processes.The massive amountofdatacollectedfromthesedevicesenablethefactoriestohavemoreefficient maintenanceandmanufacturingprocessesduringtheproduction.Ourfocusasause caseistheconveyorbeltsinfactories,whichmaysufferfromunexpectedbreakdownif oneofitscomponentswereworn-outafteralongtimeofuse.Inordertopreventthe time and money loss caused by an unintended breakdown, we want to achieve predictive maintenance,inwhichtheconditionsofthedevicesarecloselymonitoredtopredictthe needformaintenancebeforeapossiblebreakdown.

Inthismasterthesis,apredictivemaintenancemechanismisimplementedforservo motors on industrial conveyor belts.After comparing with other approaches, a one-class support-vectormachinebasedsemi-supervisedanomalydetectionalgorithmisselected: firstofall,amodelthatcapturesthecharacteristics(e.g.normalspeed-powerbe-haviours)ofthemotoristrainedusingnormalhistoricalrunningdata.Then,areal-timeabnormalbehaviordetectorforthemotoroperationisimplementedbasedonthe previousmodel. Ourexperimentresultsshowthattheimplementedmechanismcan effectivelydetectwhethertheservoisrunningnormallyornot.Whenanabnormalbe-havior is detected, a timely reminder for predictive maintenance is thrown.Additionally, asimulationisdevelopedasthedigitalreplicaofourmotor,whichisusedingenerating datafortrainingandrigoroustests.

Thepremiseofthisstudyisnotonlytopredictandnotifyapossiblemaintenance needoftheservomotors,butalsototakedecisionsforkeepingtheoperationwhile preventingfurtherharmonthemotor.Ageneticalgorithmbasedadaptivecontrolal-gorithmisproposedforthispurpose.Itselectsthemostsuitablespeedaccordingto theabnormalloadtorquesdetectedontheconveyorbelt.Thevalueoftheabnormal torquecanneverbepredictedasthesourceofitismissing.However,anapproximation canbemadethroughthedifferencebetweenthecurrentandhistoricalpowervalues. Thehistoricalpowercanberepresentedbyamathematicalmodel.Itisgeneratedby polynomialregressionanalysisfromnormaloperationdatacollectedinthepredictive maintenancephase.Suchanadaptivecontrolhelpsthemotortoavoidfurtherdamages underthesystemfaultsituation.Sinceitadjuststhespeedofthemotorintelligently, itprovidesamoreefficientprocesscomparedtoasystemthatterminatestheoperation topreventanabnormalcondition.Experimentresultsindicatethatourimplemented mechanismofpredictivemaintenanceandregression-basedadaptivecontrolleadstoa moreefficientoperationcomparedtorule-basedorpreventiveapproaches.

Zusammenfassung

DieIndustrie4.0RevolutionunddietechnischenFortschritteinderForschungzumInternetderDingebringenunsintelligentereundmehrvernetzteSensorenundGer¨atef¨ur industrielleProzesse.DieernomeDatenmenge,dievondiesenGer¨atengesammeltwerden,erm¨oglichenesFabriken,w¨ahrendderProduktioneffizientereWartungs-undFertigungsprozessedurchzuf¨uhren.UnserFokusalsAnwendungsfallsinddieFließb¨ander inFabriken,dieunterunerwartetenAusf¨allenleidenk¨onnen. UmdenZeit-undfinanziellenVerlustdurcheinenunbeabsichtigtenAusfallzuvermeiden,wollenwireine pr¨adiktiveWartungerreichen,beiderderZustandderGer¨ategenau¨uberwachtwird, umdenWartungsbedarfvoreinemm¨oglichenAusfallvorherzusagen.

IndieserMasterarbeitwirdeinpr¨adiktiverWartungsmechanismusf¨urServosaufIndustrieflließb¨andernimplementiert.NachdemVergleichmitanderenAns¨atzenwirdein”semi-supervised anomaly detection” Algorithmus basierend auf ”one class Support Vec-tor Machine” implementiert:Zun¨achst wird ein Modell, das die Eigenschaften (z.B. Normalgeschwindigkeit-Leistungsverhalten)desMotorserfasst,mitnormalenhistorischen Betriebsdatentrainiert. Anschlie˜AendwirdbasierendaufdemVorg¨angermodellein Echtzeit-Anomalverhaltensdetektorf¨urdenMotorbetriebimplementiert.UnsereVersuchsergebnissezeigen,dassderimplementierteMechanismuseffektiverkennenkann, obdasServonormall¨auft.WenneinabnormalesVerhaltenfestgestelltwird,wirdeine ErinnerungandiepR¨adiktiveWartungausgel¨ost.Zus¨atzlichwirdeineSimulationunseresMotorsentwickelt,mitderDatenf¨urTrainingundstrengeTestsgeneriertwerden.

Die Pr¨amisse dieser Studie ist es,nicht nur einen m¨oglichen Wartungsbedarf eines Servo vorherzusagen und zu melden, sondern auch Entscheidungen zu treffen, um den Betriebsablauf aufrechtzuerhalten und weitere Sch¨aden am Motor zu vermeiden.Daher wird ein aufeinemgenetischenAlgorithmusbasierenderadaptiverRegelalgorithmusvorgeschla-gen.Es w¨ahlt die beste Drehzahl entsprechend den auf dem F¨orderband erfassten abnormalenLastmomentenaus.DerWertdesabnormalenDrehmomentskannnievorhergesagtwerden,dadieUrsachedaf¨urfehlt.EineAnn¨aherungkannjedochdurchdieDifferenzzwischenaktuellerundhistorischerLeistungerfolgen.DiehistorischeLeistung kannmiteinemmathematischenModellberechnetwerden.Eswirddurchpolynomiale RegressionsanalyseausDatenimNormalbetrieberzeugt,dieinderPhasederpr¨adiktivenWartungengesammeltwurden.EinesolcheadaptiveSteuerunghilftdemMotor, weitereSch¨adenunterderSystemfehlersituationzuvermeiden.DaerdieDrehzahldes MotorsintelligenteinstellT,bietetereineneffizienterenProzessalseinSystem,dasden Betrieb beendet, um einen anormalen Zustand zu verhindern.Unsere Versuchsergebnisse deutendaraufhin,dassunserimplementierterMechanismusderpr¨adiktivenWartung undderregressionsbasiertenadaptivenSteuerungzueinemeffizienterenBetriebf¨uhrt alsregelbasierteoderpr¨aventiveAns¨atze.

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1Introduction

1.1MotivationandProblemDefinitions

The last three industrial revolutions have largely shaped today's industrial processes and human life.**The fisrt industrial revolution brought us the very first industrial machinery,** water power and steam power; the second gave birth to the electricity and assembly lines; thethirdopenedaneraofcomputersandautomation.Nowwearemovingforwardtowardsthefourthindustrialrevolution-Industry4.0(I4.0)[6].Justlikeitspredecessor, thisrevolutionhasthepowertocompletelychangeourlifebyadoptingnewconcepts liketheInternetofThings(IoT),Cyber-PhysicalSystems,cloudcomputingandmachinelearningtechniquesintheIndustrialprocesses[7].UsingCyber-PhysicalSystems, themanufacturingprocesscanbemonitoredandcontrolledremotely.ThankstoIoT, information generated during manufacturing can be collected and shared among sensors and actuators in real-time for autonomous control purposes.Furthermore, with the help ofcloudcomputingandmachinelearningtechniques,massiveinformationcanbeuploaded,storedandanalyzedonthecloudtoexploitmorevalues.

Theautonomousindustrialproductionline,includingrobots,conveyorbelts,various sensorsandotheractuators,isalreadyapowerfultoolincurrentfactories.Theywill continuetoplayanimportantroleinI4.0.Itisnotablethatintheconventionalproductionlines,therearestillverybulkyrobotsandpilesofcables.Additionally,they stillrequirehumaninvolvementbothintheproductionlinesandasthesupervisorin monitoringtheoperations,maintainingthedevicesandmakingcontroldecisionsonthe process.However,inI4.0,therewillbeflexiblerobotsthatcantakeoverjobsthatare bulkyandmonotonousforhumans;smartsensorsthatcanmonitortheproductionline automatically and communicate with each other;and devices that can predict their own maintenanceperiodandactivelyadjusttheiroperationaccordingly.Thisisexpectedto relievehumanoperatorsfromthephysicalandcognitiveload.

ThisthesisworkispartofaprojectcalledCHARIOT[8].CHARIOTisanIoTbased middlewarethatintegratesvariousservicesanddevicesthroughacquiring,storingand sharingtheirheterogeneousdata.Theprojectfocusesonapplyingthemiddlewarein I4.0concept,buildingasmartfactoryasitsusecase.Therefore,themiddlewarealso integratesaknowledgemanagementsystemthatprovidesmachinelearningbaseddata analysisandthesharingandstoringofthisdata.Additionally,theCHARIOTmiddlewareintegrateshumansinIoTallowinghumanstoaccess,communicateandcontrol sensorsandactuators.

PredictiveMaintenanceisanimportantusecasefortheCHARIOTframeworkandfor thisthesis,asitisforI4.0applications.Theadvancesinmachinelearningtechniques havetransformedtheconventionalapproach,preventivemaintenance,intopredictive maintenance[9].Thepreventivemaintenanceisperformedregularlyondevicestode-creasethelikelihoodofbreakingdown[1].Thismeans,maintenanceonlytakesplace whenthedevicereachestheendofitsmaintenanceperiod.However,consideringthe caseswhenthedeviceswerewornoutfasterorslowerthanusual,preventivemainte-nanceiseithernothelpfulornoteconomicallyefficient.Thepredictivemaintenance,on thecontrary,ismoreintelligentandefficientbecauseitcandetectsignsthatindicate possible system failures.The motivation behind predictive maintenance is to increase ef-ficiency and reduce the costs.In modern production lines,small defects can easily cause a chain reaction and paralyze the whole production line completely for weeks.This kind of break-downs will cause tremendous money loss, including the repair expenses and loss causedbythehaltofproduction[9].Makinguseofpredictivemaintenance,thefactory managerwillbeabletoschedulethebestmaintenancetimefordevicesorcomponents, thusminimizingthedowntimeandcosts.

Inthisthesis,weapplypredictivemaintenancetotheservomotorofanindustrial conveyor belt.Machine learning based anomaly detection techniques are used in our use case.Theyareimplementedinasemi-supervisedway,i.e.trainingandpredictionare separated.Inthetrainingphase,clusteringalgorithmsareusedtoidentifythechar-acteristicsofthesystem;afterward,aclassifierisimplementedbasedontheprevious clusteringresults.Morespecifically,firstofall,operationaldataarecollectedfromthe servoduringitsnormaloperation.Thedataincludethemomentarycurrent,effective voltage,angularvelocity,etc.Secondly,inthetrainingphase,usingthedatawetrain amodelthatcalculatesthedecisionboundary,i.e.aboundarythatdefinesthenormal rangeofthecharacteristics(e.g.speed-powerbehaviour)ofaservomotorunderits normaloperation.Inthisphase,thechallengesare:(1)notallclusteringalgorithms performwellconsideringthenon-circularshapeofthecollectedspeed-powerdata;(2) some algorithms train slowly if the given dataset is large;(3) the algorithm must be able to generate a model,a threshold or a dataset,to be used as a classification metric in the implementation of the classifier later.Thirdly, we use the generated model in a real-time classificationalgorithminordertodetectwhethertheservomotorisrunningnormally orthereisabnormalbehavior,forexample,abnormalfrictiononthebeltsurfaceor theonthegearofthemotor.Thechallengeherewouldbetoimplementanalgorithm thatcanmakeuseofthetrainedmodeltoclassifyadatapointasnormalorabnormal inashorttime.Ingeneral,thoughtherearealotofavailablecandidates,itisnot sure whether these algorithms can fulfill our requirements.Therefore, we implement one classsupportvectormachines(oneclassSVM),k-nearestneighbor(*k*-NN)andkernel principlecomponentanalysis(kernelPCA)inasemi-supervisedwayandcomparetheir performances in training time, classification time and classification accuracy.In the end, weselectandintegratetheoneclassSVMapproachinourconveyorbeltsystem.Since

thisprocessrequirescoveringallpossiblenormalconditionsaswellasabnormalcases, additionally, we develop a simulation as the digital replica (or “digital twin”) of our servo motorintherealsystem.Thesimulatedmotordataisusedfortrainingandrigorous testsofthesolutions.

Asanoveltyofthisthesis,weprovidearegressionbasedadaptivecontrolmechanism fortheservomotorinresponsetothenormalorabnormaloperation. Apredictive maintenancealgorithmcanprovideafastandreliablemethodtodetectthepotential anomaliesinthesystemorfluctuationsofloadtorque.However,ourgoalisnotonlyto detecttheabnormalbehaviours,butalsotoprovideanadaptivecontrolforsuchcases tokeeptheservomotorandsotheoperationrunningsmoothlyandefficiently.Therefore,weimplementanadaptivecontrolmechanismfortheservomotor.Tosaveenergy andtopreventpotentialdamage,theconveyorbeltshouldbeabletochangeitsspeed accordingtoextreme/unexpectedfluctuationsexistingintheloadtorque.Duringthe realoperation,wehavenopriorknowledgeonifsuchfluctuationsisasystemfaultor ifitisanormalloadtorquefluctuation.Ifitistheformer,thereisnoinformationon whichcomponentisthemaincauseofsuchanabnormalbehaviorinacomplexsystem. Therefore,agenericalgorithmforbothsituationsshouldbeimplementedtotakeadaptivedecisionshandlingallcases.Forthepurpose,weimplementageneticalgorithm (GA)asouradaptivespeedcontrol.

Thisalgorithmisanextensionofourpredictivemaintenancealgorithm.Thepredictionprovidesanotificationtotheuseraboutapossiblemaintenanceneedwhereasthe adaptivecontroltriestoprotectboththeoperationandthemotorbystillrunningitin aprotectivemode.Thegeneticalgorithmisselectedasouradaptivecontrolscheduler. Thehistoryspeedandvelocityvaluesareusedtoconstructchromosomes.Ineachchromosome,aspeedinterval,aswellastheupperandlowerboundofthecorresponding powerofthespeedinterval,areincluded.Consideringthefactthatthespeedinterval sizecanvary,itwouldbeunrealistictomaintainalargehistoryspeed-powerdatabase thatcoversallthespeed-powerreadingsweneed.Instead,amathematicalmodelis builtbypolynomialregression.Theregressionidentifiestherelations($P=f(v, T)$) among power (P), velocity(v) and load torque(T) from the history data collected during thepredictivemaintenancephase.Usingthismathematicalmodel,givenvelocityand torque,thepowerundernormalconditionscanbecalculated.Inthisway,ourcontrol algorithm is able to calculate the deviations from the normal behavior of the load torque throughvelocityandpowerreadings.Additionally,aproportionalˆaintegralˆaderivative (PID)errorcalculationisimplementedtotriggertheGA.PIDalgorithm,although mainlyusedincontrol,caneffectivelycalculatethedeviationsconsideringtheeffectsof thecurrent,pastandfutureerrorofthesystem[10].ThePIDerrorcalculationaccepts real-timepowerreadingandhistorypowerreadingasinputs;onlywhenthecalculated PIDerrorexceedsthepre-definedthreshold,GAistriggered.Inthisway,unnecessary computationoverheadcanbeavoided.Ingeneral,throughfitnesscriteria,ouradaptivecontrolalgorithmisabletodecideontheoperatingspeedinabnormalcases,while

keepingtheuserrequestedspeedforthemotorinnormalcases.Forexample,ifthe load torque increases to a critical level, the algorithm should be able to decelerate motor tobalancethepowerdrained,inthemeantimeprotectingtheoperationspeedwhen possible.Lastbutnotleast,theeffectsofdifferentintervalsizeanddifferentfitness functionweightsontheperformanceoftheadaptivecontrolalgorithmarecompared; the one with the best performance are further compared with the rule-based mechanism.

Resultssuggestthatouralgorithmsignificantlyoutperformstherule-basedone.

1.2Objectives

The principal objective of this thesis is to develop the predictive maintenance and adaptivecontrolmechanismfortheconveyorbelt.Additionally,aRepresentationalState Transferstyle(RESTful)basedIoTAPIisimplementedtointegratethesetwomecha-nisms to the CHARIOT project middleware.The major tasks are separated into smaller goalsandaregivenasfollows:

1.Detailedresearchonpredictivemaintenanceandadaptivecontrol.

2.Implementation of the hardware setups for the experiments,including the integra-tion of the servo motor with the conveyor belt,the integration of micro-controllers (RaspberryPi)withtheservomotoraswellastheintegrationoftherobotarm withtheconveyorbelt.

3.Collectionoftheservomotoroperationdata(i.e.velocity,power,current,etc.) whenrunningunderthenormalconditions.Constructingasimulationmotoras a“digitaltwin”oftherealmotorforsyntheticallygeneratingdatacoveringall conditionsandforrigoroustests.

4.Testingseveralanomalydetectionalgorithmstofindthebestperformergiventhe servomotordata,andtotrainamodelthatcancapturethecharacteristicsofthe motor.Lateron,usingthetrainedmodeltoimplementthereal-timeanomaly detector(i.e.abnormalbehaviorsofthemotor).

5.Conducting regression analysis to build a speed-torque-power mathematical model. Implementing and tuning a GA to fit our requirements.Implementing a PID error calculation to trigger the GA for new speed selection.Combining the mathematical model,GAalgorithmandPIDerrorcalculationtoconstructtheadaptivecontrol algorithm.

6.ImplementinganinterfacefordataexchangewiththeIoTbasedmiddlewareon thecloud.

7.TestingalgorithmsandcomponentsimplementedinSection3.3,Section3.4and Section3.5.

1.3Contributions

Asthisthesisisapplicationbased,themajorcontributionofthisthesisistocombine existingmethodsandalgorithmstobuildacompletesolutionforthisspecificusecase: predictivemaintenanceandadaptivecontrolfortheservomotorinindustrialconveyor belts.Morespecifically,themaincontributionofthisthesisconsistsof:

1.Implementing a semi-supervised anomaly detection algorithm that supports offline trainingandonline(real-time)predictionoftheabnormalbehaviorofaservo motor.

2.ProposingandimplementingapolynomialregressionandGAbasedadaptivecon-trolmechanismfortheservomotorthatdecidesonnewprocessmodestoprotect boththeoperationandthemotorinsuchabnormalcases.

3.Integrating the solution on a middleware's knowledge management system through aneasy-to-useRESTfulAPIthatprovidesnotonlyMySQLdatabaseaccess,but alsomachinelearningbaseddataanalysis. Thisallowstheotherdevicesand applicationsinthefactory(includingouradaptivecontrolmechanism)toaccess thedatageneratedafterMLoperations.

4.Applicationoftheknowledgemanagementsystemincorporatingourpredictive maintenanceandadaptivecontrolsolutionsonarealconveyorbeltsystemandits"digitaltwin"f orreal-timeperformance.

1.4Outline

Thisthesisisseparatedinto6chapters.Theremainingchaptersaftertheintroduction areorganizedasfollows:

Chapter2reviewsthestate-of-arttechnologiesforpredictivemaintenance,adaptive controlandknowledge-basedIoTsystems.Differentexistingapproachesareanalyzed, revealingthefieldsthatlackresearchandthepotentialsolutionsforourproblems.

Chapter3describestheoverallstructureofthesystem,aswellastheapproaches andimplementationoffourmajorparts.Firstofall,thecollectionofdataandthe implementationofthesimulationmotorisintroduced.Next,inpredictivemaintenance part,semi-supervisedlearningapproachesformotorcharacteristic"fingerprint"genera-tionandreal-timeanomalydetectionarepresented.Afterthat,intheadaptivecontrol part,theimplementationoftheregressionandtheGAbasedadaptivecontrolarede-scribed.Lastbutnotleast,theRESTfulAPIforMySQLdatabaseaccessandmachine learningbaseddataanalysisarepresented.

Chapter4describestheprotocolsofevaluationandevaluationresults.Detailsabout

the experiment settings and system architecture, how maintenance algorithms and adaptivecontrolalgorithmareoptimizedandtestedaredescribed.Theresultsoftheexperimentsarealsoanalyzedanddiscussed.

Chapter5summarizesthethesis,inwhichtheproblemsaredescribed,theapproaches andimplementationareoutlined,andanoutlookaboutfutureworkisgiven.

2RelatedWork

Inthischapter,thestate-of-the-arttechniquesinpredictivemaintenanceandadaptive control area are presented.In the predictive maintenance part, first of all, the traditional methods in maintenance are introduced.Next, the new maintenance method:predictive maintenance,especiallythecasesinwhichitcombineswithmachinelearning,isstud-ied.Afterthat,basedonourusecase,theanomalydetectiontechniquesforpredictive maintenancearediscussed.Asforadaptivecontrolpart,firstofall,thecommonlyused adaptivecontrolconceptandvariousapproachesarepresented.Then,afterdeciding togoforthegeneticalgorithm,thestate-of-artstechniquesofprocessesinthegenetic algorithmareresearched.Finally,regressionbasedcontroltechniquesarestudiedto figureouthowitmighthelpinourusecase.

2.1PredictiveMaintenance

2.1.1TraditionalMethodsinMaintenance

Tobetterunderstandpredictivemaintenance,firstofall,conventionalmaintenance methodsshouldbestudied.Twotypesoftraditionalmaintenancemethodsareused inindustry:run-to-failureandpreventivemaintenance[1].

Run-to-failureisalmostthemostprimitivemaintenancemethod. Itissimpleand straightforward:runamachineoracomponentuntilitbreaksdown;otherwise,don't fix it.As a purely reactive approach, run-to-failure does not take any precautions, there-foreitcanbeveryexpensive:themaintenancedepartmentneedstomaintainextensive sparepartsincaseanycomponentfails.Inpractice,apurerun-to-failureapproachis hardtoapply,sinceinordertomakethesystemrun,basicpreventivetechniquessuch aslubrication,machineadjustments,etc.willbeperformedeveninarun-to-failureen-vironment.Inmodernindustry,run-to-failureisthesituationthatmostmanufacturers wanttoavoid.However,run-to-failurecanbeveryusefulfortestingpurposes,forex-ample,productdurabilitytests.

Despite many definitions of preventive maintenance,it is always considered time-driven; therefore,itisalsocalledperiodicmaintenanceinsomeresearchpapers[11].Inmostof the preventive maintenance, maintenance (inspection, replacement, etc.)is usually done visuallyandphysicallyinsteadofusingsensors.Furthermore,amaintenanceisusually scheduled based on the mean-time-to-failure (MTTF, as shown in Figure 2.1) and mean-time-between-failures(MTBF)statistics.TheestimationofMTTForMTBFishighly

dependingonhistoricalrecordsandexperiences.Althoughpreventivemaintenanceis still widely used,it is being increasingly regarded as defective and unreliable [1].In [12], testsareconductedtocalculatethetime-to-failureusingidenticalsystemsunderidenti-cal conditions.Results indicate that the time-to-failure is very different from one system toanother.Thisprovestheineffectivenessofpreventivemaintenancesinceitishardto knowwhenthesystemisgoingtofailbasedonperiodicmaintenance[11],whichwill furtherleadtoinefficientcomponentreplacementorcatastrophicsystembreakdown.

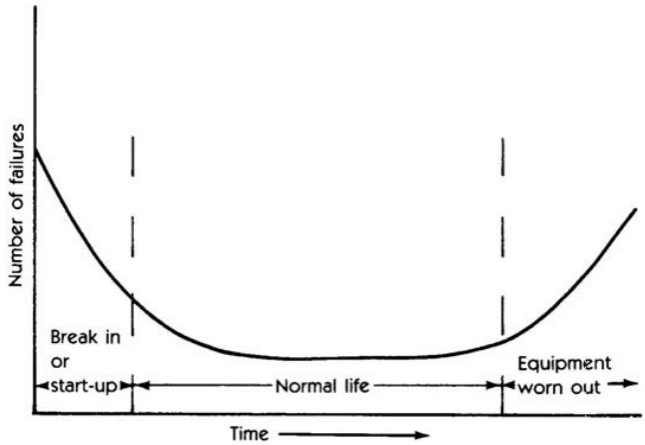


Figure2.1:TypicalMTTFbathtubcurve[1]

2.1.2PredictiveMaintenanceandMachineLearning

Predictive maintenance is a condition-driven preventive maintenance method.Compared to the conventional preventive maintenance, rather than depend on historical MTTF and MTBFestimationstoschedulemaintenance,itdirectlymonitorstheconditionsofthe devicestodecidewhethermaintenanceisneeded[1].Therefore,inthiscase,themaxi-mumintervalbetweenrepairsandtheminimumdowntimecanbeachieved.

Variouspredictivemaintenancemethodscanbeclassifiedintothreemaincategories: modelbased,case(rule)basedanddata-drivenbased[13].Overtheyears,predictive maintenancehasevolved.Nowadays,data-drivenbasedmethodshavebecomethemost popularones,thankstotheadvancesinmachinelearningtechniques.Machinelearning techniques also increase the feasibility and availability of predictive maintenance [14];as itisnowpossibletoapplypredictivemaintenancetofieldsthatarehardtomaintain previously[15] [16].

Predictivemaintenanceinindustryincludesdifferentaspects,forexample,calculation oftheRemainingUsefulLife(RUL)[17],conditionmonitoringtopreventpremature failures[1],etc.MachineLearningtechniques,suchasclustering,classification,regres-sionandneuralnetworks,arewidelyappliedtothethesearea[16][14][18][19]andare

proventobeeffective.

Inthisthesis,toexploitvaluablereal-timeinformationfrommassivedataobtained fromtheservomotoranditsperipheralsensors,wewanttoachievereal-timepredictive maintenance,i.e.todecidewhethermaintenanceisneededbasedonthecurrentsystem conditions.Thisfallsintothedomainofanomalydetection.Despitethecloserelations betweenmachinelearningbasedanomalydetectionandpredictivemaintenance,there isnoresearchdirectlycombiningthosetwoconcepts.Inanomalydetection,theclosest applicationtopredictivemaintenanceisfaultandfailuredetection.Therefore,inthe nextsubsection,differentcategoriesofanomalydetection,aswellasthepotentialof applyingthemtopredictivemaintenance,areresearched.

2.1.3AnomalyDetectionandItsDifferentCategories

Anomalydetectionoroutlierdetectionreferstotheproblemoffindingdatapointsthat do not follow the expected behavior.These abnormal data points are usually considered as anomalies, exceptions, outliers or contaminants in different applications [20].Another similarbutmorespecificconceptinmachinelearningisNoveltyDetection.Itrefersto thesituationthatthetrainingdataareall”normal”,i.e.notcontaininganyoutliers; butwearemoreinterestedindetectingwhetheranewobservationisanoutlier[21]. Underthiscircumstance,anoutlierisalsocalledanovelty.

Anomaly detection can be done using supervised, semi-supervised or unsupervised learn-ingtechniques.Supervisedanomalydetectionisonlyapplicabletoscenariosthatcan guarantee the availability of both classes labeled as ”normal” and ”abnormal”.A typical approach, in this case, is to build a model using classification algorithms.Then the novel data are compared to the model to see which class they belong to.However,in anomaly detectionproblems,theabnormaldatainstancesareusuallymuchlessthanthenormal ones.Thisissuehasbeenaddressedin[22],[23],[24]etc.Toresolvetheissues,different methodshavebeenintroducedtogenerateartificialanomaliestoobtainareasonably balancedlabeledtrainingdataset[25][26].However,inourcase,itwouldbehardfor ustoadoptthismethodsincewehavelimitedinformationabouttheabnormaldata.

Unsupervisedanomalydetectionismorewidelyusedasitavoidstheproblemmen-tionedabovebyimplicitlyassumingtheamountofnormaldataarewaymorethanthe abnormaldatainthetrainingset.Thesetechniqueshavebeensuccessfullyappliedto intrusiondetection,frauddetection,dataleakageprevention,medicalapplications,etc. [27].However,oneshouldnoticethat,inallthesesuccessfulapplications,trainingdata arecollectedbeforehand.Unsupervisedmethodsmaysufferfromslowtrainingspeedif theoriginaltrainingdatasetisbigandthemodelneedstobere-trainedwhennewdata areadded.Consideringthedynamicnatureofreal-timeanomalydetection(wherenew data come in all the time), pure unsupervised techniques are not suitable for our scenario.

In[9],theresearchersuseclusteringalgorithmstoacquirethecharacteristic”finger-

print”(e.g.power-torquegraph,power-shaftaccelerationgraph,etc.)oftheservo motorduringitswholeaccelerationanddecelerationprocessforpredictivemaintenance purpose.Thepaperalsosuggestsitisexpectedthattheservomotorwillremainthis”fingerprint”inthef utureoperations.Therefore,anomaliescanbedetectedusingthe generated”fingerprint”whenthegenerateddatapointsareoutsidetheclusters.Thisis actuallysemi-supervisedanomalydetection,inwhichthetypicalapproachistotraina model for the normal behavior class first,and then use the model to detect anomalies in thetestdataset[20].Thismethodisparticularlysuitableforourscenarioinpredictive maintenance, because first of all, it is difficult to generate abnormal behaviours but much easiertogeneratenormalbehavior;Secondly,insemi-unsupervisedmethod,trainingof the model only needs to be done once and the anomaly detection for new data point will beperformedrecursively;asthepredictionforasingledatapointismuchfasterthan themodeltraining,theseparationofthesetwopartshasguaranteedashortreaction time,whichisessentialinreal-timeanomalydetection.

2.1.4AnomalyDetectionTechniques

In the last part, we have discussed the different categories of anomaly detection.Another similarconceptthatworthmentioningisnoveltydetection.Thenatureofimbalanced trainingdataclassdistribution(i.e.onlynormaldata,nooutlier)andthefocusonnew observations in novelty detection,resemble the assumptions in semi-supervised anomaly detectionapproaches.Therefore,wedon’tdistinguishanomalydetectionandnovelty detectioninthisthesiswork.Inthispart,state-of-artnoveltydetectiontechniquesare discussedandsuitabletechniquesareconsideredforourimplementations.

Variousnoveltydetectiontechniquescanbecategorizedintoprobabilistic,distance-based,reconstruction-based,domain-based,etc.Amongthem,thek-nearestneighbor approaches(*k*-NN)aredistancebasedandarewidelyusedfornoveltydetection.*k*-NN is based on the assumption that normal data points all have close neighbors.Therefore if adatapointisfarawayfromitsneighbors,itisregardedasanoutlier[28].Specifically, toapplyittobuildasemi-supervisedmodel,oneshouldfirstdeterminethedistance thresholdusingthearrayobtainedfromcalculatingthedistancesbetweeneverydata pointandinthetrainingdatasetanditsk-nearestneighbors.Thisthresholdislater usedasthenoveltymeasureforprediction.Whilethenoveltyscorecanbeeitherthe distancetothe*k*thnearestneighbor[29],ortheaveragedistancetoallthek-nearest neighbors [27],or the median distance to the k-nearest neighbor [30].For each new data point,ifthenoveltyscoreexceedsthethreshold,thenitisregardedasanovelty.This methodiseasytoimplement,however,sincedistancesneedtobecalculatedforevery datapoint,itisalsocomputationallyintensiveandtime-consumingifthetrainingsetis large.

Despiteitssimplicity,beinganon-parametricmethod,itisoftensuccessfulinclassifica-tionsituationswherethedecisionboundaryisveryirregular.Therefore,thek-nearest neighboralgorithmwillbeasuitablecandidateforoursemi-supervisedanomalydetec-

tionalgorithms.

Oneclasssupportvectormachines(SVM)isadomain-basednoveltydetectionalgorithmthatisalsopopular.Originally,supportvectormachineisusedtogeneratea non-lineardecisionboundaryamongdifferentclassesbyprojectingthedatatoahigher dimension.First proposed by Sch¨olkopf et al.[31], one class SVM use a similar approach toformadecisionboundarytightlyaroundthetrainingdata.Morespecifically,usinga kernelfunction,theboundaryisdefinedbyahyperplane,whichcaneffectivelyseparate the training data points from the origin in the feature space with the maximum distance [32].While in another approach, the Support Vector Data Description (SVDD) from Tax andDuin[33],thedecisionboundaryisdefinedbyahyperspherethatcanencompass mostofthetrainingdatawiththeminimumvolume.However,thesecondapproach faceissuessincethedataarenotsphericallydistributedinmostofthecases.Several other papers [34], [35] proposed their solutions to mitigate this problem.Later proposed noveltydetectionalgorithmsinSVMareaaremostlybasedoneitheroneclassSVM orSVDD.Inourcase,consideringthegeneralcase,thedatawillhardlybespherically distributedeveninahigherdimension,therefore,oneclassSVMwillbeabetterchoice forthesemi-supervisedimplementation.Morespecifically,inthetrainingphase,the decisionboundaryiscalculated;lateron,inthepredictionphase,thenoveltymeasure willbewhetheradatapointiswithinoroutsidetheboundary.

Principle Component Analysis (PCA) is a reconstruction based approach.It is a method using orthogonal transformation to reduce the dimensions of the data while maintaining its original information as much as possible.However, people discovered that it also provided anewviewpointtodistinguishnoveltyinthetransformedspace.HeikoHoffmann in[36]proposedamethodtoapplykernelPrincipleComponentAnalysis(kernelPCA) tonoveltydetection.UsingRadialBasisFunction(RBF)kernel,thestandardPCAis mappedtoaninfinite-dimensionalfeaturespace.Here,aprincipalcomponentsanalysis isperformed,andthereconstructionerrorofPCAisusedasthenoveltymeasure.AccordingtoHoffmann,thisapproachhasbetterperformancethanoneclassSVMwhen appliedtohand-written-digitandbreastcancerdatasets.Therefore,thismethodwill beagoodcandidateforourimplementation.Onecanimplementthesemi-supervised methodbyfirsttransformingthedatafromtheoriginalfeaturespacetotheinfinite-dimensionalspaceandviceversa;thenthereconstructionerrorinthelastprocesscan beviewedasanoveltymeasureinprediction.

Therearealsootherapproaches,forexample,K-means(distance-based),Replicator NeuralNetwork(reconstruction-based),autoencoder(reconstruction-based),etc.Some ofthemareeithernotsuitablefornon-sphericaldata[37]ornotfastenoughforreal-timeprediction.Astherearealotofavailableapproaches,takingthedisadvantages andadvantagesintoaccount,wechosethreepopularalgorithmsintermsofnoveltyand anomalydetection,i.e.oneclassSVM,*k*-NNandkernelPCAasoursemi-supervised real-time anomaly detection methods.We will further compare these algorithms to figure

outthebestone.

2.2AdaptiveControl

2.2.1AdaptiveControlofMotors

Therealreadyexistsmuchresearchfortheadaptivecontrolofmotors.Theseadaptive control methods are all targeting at improving speed regulation performance or position controlaccuracyofthemotorcontrollers.Atthefirstbeginning,the”adaptivecontrol”conceptispropo sedtosolvetheparametricuncertaintyproblems,suchasextranoise, variationaloperatingenvironment,wearofcomponents,etc.inmotoroperations[38]. It is worth noticing that, this specific adaptive control concept, although not serving the samepurposeasourusecase,istryingtodealwithasimilarsituation(extranoises, disturbance, etc.).Therefore, the research in this area can provide us with insights about howtodesignandimplementourownalgorithm.

Adaptivecontrolreferstoestimatingthemotorparametersonlineusingthesystem feedback[39].Then,theestimatedparameterswillbeusedtocomputethecontrolsig-nalsforthemotorsothatthemotorcanreachthedesiredperformance.Onekindof approachesforadaptivecontrolisartificialintelligence(AI)based.Therearealotof researchesonusingfuzzycontrolorfuzzyrulestotunethePIDcontrollerparameters asin[40], [41]and[42].Whilein[43],aneuralnetworkisconstructedtoacquirethe motorcontrollergainsforaparallelrobotmanipulator.AIbasedadaptivecontrolhas proventobeeffective,however,itrequirescollectingdatathatcontainsysteminforma-tion beforehand and training a model according to it;therefore it might not be adaptive enoughwhentheenvironmentischangingdrastically.

Anotherwayofdoingadaptivecontrolisusingmeta-heuristicmethods.InFisteret al. [44],variousofmeta-heuristicmethodswereusedtotunethePIDparametersofa SCARArobotoff-line,inwhichParticleSwarmOptimization(PSO)stoodouttobe thebestalgorithm.InVillarreal-Cervantesetal.[45],differentvariantsofDifferential EvolutionalgorithmswerealsousedtotunetheparametersofaPIDcontrolleroff-line. Furthermore, meta-heuristic methods can also be used online.In Lin et al.[46], Genetic Algorithm was used to tune the parameters of a PID controller on a linear induction mo-tor online to achieve the minimal position control error.Although all the meta-heuristic methods mentioned above achieve good performance, [38] also mentioned that one draw-back the users need to bear is the long convergence time.This might be a problem when theuserswanttoupdatethecontrolsignalinareal-timemanner.

Despite the fact the meta-heuristic methods have convergence issues, it is hard to neglect that,mega-heuristicalgorithmsplayanimportantpartinthecurrentadaptivecontrol researches.Henceitwouldbeagoodideatoextendthecurrentresearchfromonline motorcontrollerparametertuningtoourusecase.Althoughin[38],experimentresults

showthatthegeneticalgorithmonlyhasaverageperformance,asthemeta-heuristic algorithmwiththelongesthistory,italsohasthemostabundantresources.Therefore itwillbethemainresearchfocusinthenextpart.

2.2.2GeneticAlgorithm

Thegeneticalgorithmisameta-heuristicsearchapproach.FirstproposedbyJohnHol-land [47] in the 1960s, the genetic algorithm has been a popular search and optimization methodduringthelastdecades.ThealgorithmisinspiredbyDarwinianevolutionary theories,inwhichseveralrulesarefollowed:insidethepopulationofaspecies,individ-ualscompeteforlimitedresourcesandthusmaintainthesamepopulation;influenced bytheenvironment,thepopulationchangegraduallyfromgenerationstogenerations viaselectionandreproduction;themostimportantruleis,thefittestindividualsfor theenvironmentsurvive.Theprocessofnaturalselectionguaranteesrandomnessofthe possiblesolutionsandthethoroughnessofthesearch,thereforethegeneticalgorithm hasbeenwidelyusedoncomplexreal-worldapplications,suchasdecisionmaking,data mining,roboticsetc.Anexplanatorygraphforthegeneticalgorithmispresentedas Figure2.2.

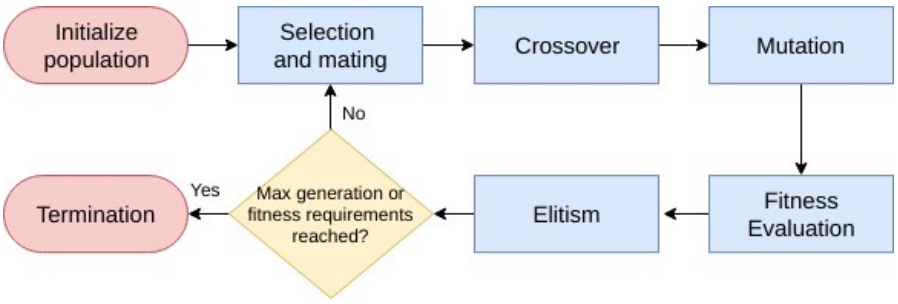


Figure2.2:AstandardGeneticAlgorithmprocess

Thetypicalgeneticalgorithmmethodologyincludesinitialization,selection,crossover, mutation,elitismandtermination.Amongthem,theselection,crossoverandmutation arethemostimportantprocedureshatgeneratevariationinthepopulation.

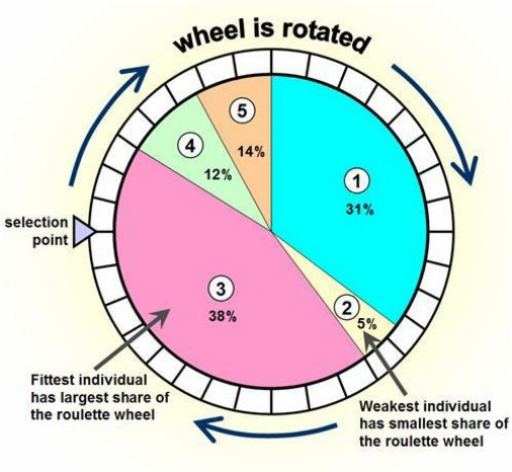
2.2.2.1EncodingandInitialization

Inordertousethegeneticalgorithm,firstofall,theproperrepresentationmethod forachromosomeshouldbeselected.Thisrepresentationmethodistermedencoding. Encodingnotonlydecideshowtheproblemisresolvedinthegeneticalgorithmbut alsodecideswhichgeneticoperatorstochoosefrom.Usually,thechromosomeofan individualconsistsofasetofgenes.Theformofthegenescanbebinarybits,integer numbers, floating point numbers, matrices, as well as symbols (e.g.Latin alphabets).By far,themostpopularencodingmethodsarebinarydigitsandfloatingpointnumbers. As[48]suggests,oneshoulduse theencodingmethodthatisthemostsuitablefor theproblem,thendevisethegeneticalgorithmaccordingly.Thebinarycodingisthe

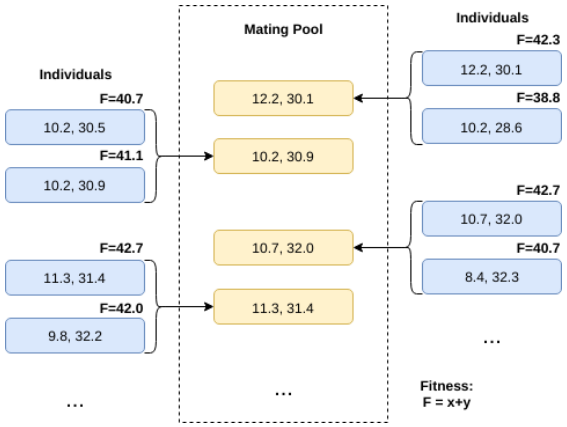
mostcommonlyusedencodingmethod,mostlybecausethefieldhasbeensufficiently researched.However,binaryencodingcanbeunnaturalformanyproblems.Inorder touseit,someadjustmentoftenneedstobedoneinadvance. Ontheotherhand, floatingpointnumberencodingismorenatural.Moreover,forproblemsthatrequire accurateresentation,suchasreal-worldoptimizationproblems,usingbinarydigit encodingwillbedifficult.Asforinitialization,thesizeofthepopulationcannotbetoo smallinordertoguaranteetheeffectivenessofthealgorithm.Moreover,individualsare generatedrandomlytocoverthewholesearchspace[49].

2.2.2.2Selection

Duringthealgorithm,inordertoproducesuccessivegenerations,parentsareselected togivebirthtooffspring. Individualsneedtogothroughfitness-basedselection,in whichanindividualwithahigherfitnesswillbemorelikelytobeselected.Popular methodsinthisstageincluderoulettewheelselection,tournamentselection,etc[50]. Roulette Wheel Selection assign different probabilities to the individuals according to its normalized fitness; therefore, the larger fitness, the larger chance.An extra conversion is required for the genetic algorithm that considers the minimum fitness as the best.While intournamentselection,acertainamount(givenbytheuser)ofcandidatescompete witheachotherandthebesthavetherighttobreed. Thismethodguaranteesthe worstindividualnevergetsthechancetoreproduce.Moreover,itisbothapplicable forminimumfitnessormaximumfitnessapproaches.Themechanismsofroulettewheel selectionandtournamentselectionareshowninFigure2.3.



(a)Theroulettewheelselection[51]



(b)Thetournamentselectionwhenthecandi- datenumber=2

Figure2.3:Comparisonofroulettewheelselectionandtournamentselection

2.2.2.3Crossover

Thenextstageiscrossover.Incrossover,chromosomesbetweenparentsrecombineto createnewdescendants. Traditionalapproachesforcrossoverinbinaryvaluedchro-mosomesincludeone-pointcrossover,two-pointcrossoveranduniformcrossover[52]. One-point crossoversimplyswapsbitsondifferent sidesof thecrossoverpointfrompar-ents;whiletwopointscrossoverswapbitsbetweentwocrossoverpoints.Thisapproach canbefurtherextendedtok-pointcrossoverforanygivenpositiveintegerkandcorre-sponding k crossover points.In uniform crossover, each bit is independently chosen from eitherparentwiththesameprobability.Real-valuedapproachesuseslightlydifferent approaches.Discretecrossoverissimilartouniformcrossover,butinsteadofexchang-ingbits,thismethodexchangesthecorrespondingfloatingpointnumbersonthesame locationofthechromosomes.Thismethodbecomeslesshelpfulingeneratingvarietyif theamountoffloatingpointnumbersislimitedinthechromosome.Abetterapproach forfloatingpointvaluedchromosomesistheintermediatecrossover.Itgeneratesvalues foroffspringaroundandbetweenthevariablevaluesoftheparents[53].Offspringare producedbasedonthefollowingprinciple:

$$V\,ar_{O\,i} = V\,ar_{P_1} + \alpha_i \cdot (V\,ar_{P_2\,i} - V\,ar_{P_1\,i}) \tag{2.1}$$

Where α_i is a scaling factor uniformly chosen in the interval of [-d, 1+d] for each variable. Agoodchoicefordis0.25togenerateenoughvariety.Asimilarmethodistheline crossover,howeveronlyoneαischosenforadescendant:

$$V\,ar_{O_1\,i} = V\,ar_{P_1\,i} + \alpha_{O_1} \cdot (V\,ar_{P_2\,i} - V\,ar_{P_1\,i}) \tag{2.2}$$

$$V\,ar_{O_2\,i} = V\,ar_{P_1\,i} + \alpha_{O_2} \cdot (V\,ar_{P_2\,i} - V\,ar_{P_1\,i}) \tag{2.3}$$

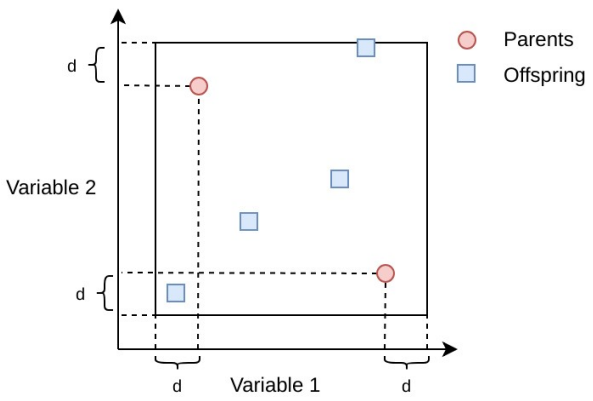
Anillustrationfortheintermediaterecombinationandlinerecombinationisshownin Figure2.4.

2.2.2.4Mutation

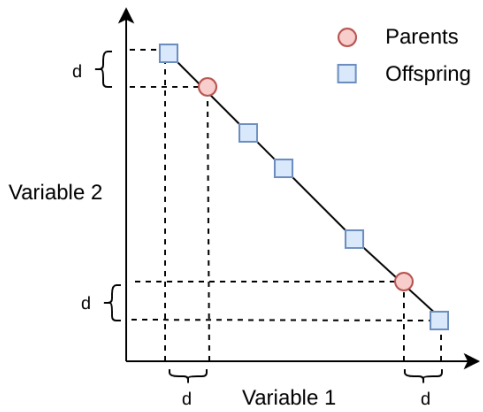
Thenextstage,mutation,workswithcrossovertogeneratemorevarietyinthepopu-lation.Atypicalmethodforbinarychromosomeisflipbit[54],i.e.invertthebitsat agivenprobability.However,thismethodisnotsuitableforfloatingpoint,giventhe factthatfordecimalfloatingpointnumber,eachbithas10alternativesintotal.Avail-able methods for floating point chromosomes are uniform mutation,Gaussian mutation, addingasmallnumber[54]etc.InuniformandGaussianmutation,anupperbound andalowerboundareselected,andthechosengeneisreplacedbyanewnumberin the given ranges,either using uniform or Gaussian distribution.Adding a small number actuallyturnsouttobeamorepopularmethod[55].Anormallydistributedrandom numberisaddedtothegene,usingthefollowingformula:

$$V\,ar_o = V\,ar_i + \sigma \cdot N(0, 1) \tag{2.4}$$

Where$sigma$isthestandarddeviationofthenormaldistribution,and$N(0, 1)$isthe standardnormaldistribution,withmean=0andvariance=1.



(a)Possibleareaoftheoffspringafterintermediate recombination(2dimensional)



(b)Possiblerangeoftheoffspringafter linerecombination(2dimensional)

Figure2.4:Comparisonofintermediaterecombinationandlinerecombination

2.2.2.5Fitnessfunction

Thecompositionofthefitnessfunctionisanessentialpartofthegeneticalgorithm,it determinestheperformanceofthealgorithm.Usually,thefitnessfunctionisdefinedby theuser.Astandardformofafitnessfunctionis:

$$F = w_1 \cdot f_1 + w_2 \cdot f_2 + ... + w_n \cdot f_n \quad (2.5)$$

Where$w_i$areweightsand$f_i$arethevariablesthatneedtobetakenintoaccount[56]. Thegeneralruletoconstructaproperfitnessfunctionisthatitshouldfaithfullyreflect thevaluesofthechromosomesanditshouldbeabletodrivethepopulationevolveto thedesireddirection.

2.2.2.6Elitism

Inordertoguaranteethatthealgorithmwillalwaysreservethefittestindividualsof thepopulation,asimplemechanismcalledelitismisintroduced[49]. Inthenormal operationsinageneticalgorithm,withoutelitism,thepopulationcaneasilylosethe bestindividual,becausecrossoverandmutationcannotguaranteethatthebestparents canalwaysbreedthebestoffspring. Theelitismsimplycopiesthefittestindividual toexternalstoragespaceandlateronthe individualiscopiedbackintacttothenext generation. Thesideeffectofelitismisthepossibilityofleadingthepopulationto prematureconvergence,i.e.thepopulationisdominatedbyasuperindividual.

2.2.2.7Terminationcriteria

Inthegeneticalgorithm,theterminationcriteriaareusuallydefinedas[49]:

•Asolutioninthepopulationhasfulfilledtheminimumrequirement

- •Maximumgenerationlimitreached
- •Budget(computationtime/money)reached
- •Optimalfitnessreachedorfurtheriterationscannotprovidebetterresults(the changeoffitnessissmallerthanapre-definedthreshold)

Differentterminationtechniquescanbecombinedtobalancespeedandperformance.

2.2.2.8Conclusion

In general, in this thesis, as we are using the genetic algorithm for parameter optimization tobalancepowerconsumptionandprocessingtime,weneedapreciserepresentationof thechromosome.Therefore,inthefirstplace,floatingpointencodingshouldbemore suitable.Forcrossover,theintermediaterecombinationischosentomakethemaximal useofthecontinuouscharacteristicoffloatingpointnumberssothatmoreaccurate crossoverscanbeachieved. Asformutation,smallnumbermutationisadoptedto ensurethemutationcangeneratevariety,butmeanwhilenotchangingthechromosome toomuch.Moreover,elitismisalsoappliedtopreservethefittestindividualinevery generation.Furtherdetailedimplementationsofthegeneticalgorithmareelaborated in Section3.4.2.

2.2.3RegressionandRegressionBasedControlMethods

Inthispart,studiesaboutregressionandregressionbasedcontrolmethodsaredoneto figure out how the regression approach will help us in the implementation of the adaptive controlmechanism.

Regressionanalysisisusuallyusedtoestimatetherelationshipsbetweenvariables. Widelyusedinengineering,regressionanalysishasbecomeanindispensabletoolin modelingandsimulations.Itcanbecategorizedintoparametricandnon-parametric regression[57].Inparametricregression,theregressionfunctionisdefinedbyafinite numberofparametersthatcanbederivedfromthetrainingdata;whilenon-parametric regressionreferstothescenariowhereaspecificregressionfunctionisabsent;either therearesetoffunctions,possiblyinfinite-dimensional;orthedatathemselvesserve asamodelaswellasestimatesofthemodel. Inparametricregression,approaches includelinearregression,polynomialregression,logarithmregression,exponentialre-gression,etc. Whileinnon-parametricregression[58],techniquesincluderegression trees,Non-parametricMultiplicativeRegression,kernelregression,Gaussianprocessre-gression,evenk-nearestneighborregression.

Therearetwotypesofcontrolthathighlyrelatestoregression.Oneisthetraditional model-basedpredictivecontrol,inwhichresearchersfirstusedstatisticalmethodssuch asregressiontobuildmathematicalmodelsfordynamicalsystemsfrommeasureddata; then,theconstructedmathmodelisusedtocalculatethecontrolsignals.Thesecond

oneiscalledmodel-freepredictivecontrol,inwhichpeopleusethehistoricalrecordsof inputsandoutputstoperformregression,andthendirectlypredictfutureinputsand outputs[59].Inordertogetgoodpredictionresults,alargeenoughtrainingdataset mustbemaintained.Lietal.[59]proposedamodel-freepredictivecontrolmethod fornon-linearsystemusingpolynomialregression.Luoetal.[60]proposedak-nearest Neighborregressionbasedmethodtopredicttheoutputofthesemiconductormanu-facturingprocessbasedonalargeamountofhistoricaldata.Ingeneral,theproblems withthefirstapproachis,firstofall,buildingamathematicalmodelformotorscanbe trickysincealotofparametersneedtobeconsidered;however,whenpredicting,the firstapproachconsumeslesscomputationalpowerthanthesecondapproach.Whilefor thesecondapproach,thetrainingiseasy;however,inordertocoverallthesituations, alargedatasetneedstobemaintained;additionally,onlineregressionmaytakelonger time.Inourusecase,sincetherelationsbetweenspeed,torqueandpowerarenotvery complicated,wecandirectlyusethefirstapproachtoobtainthemathematicalmodel whichcanbedirectlyusedinthefuturecomputation.

2.3Conclusion

Tobrieflysummarizetherelatedworks,itisnotablethatthoughalotofresearches existinbothpredictivemaintenanceandmachinelearningbasedanomalydetection, somehowresearchthatdirectlycombinesbothisstillmissing.Nevertheless,fromthe existingresearch,wealreadyseethepotentialofanomalydetectiontechniquesgetting good performance in the predictive maintenance domain.Three algorithms,namely one classSVM,*k*-NNandkernelPCA,areidentifiedaspromisinginourusecase.However, it is not easy to qualitatively select an optimal one.Therefore, all of the three algorithms willlaterbeimplementedandcomparedwitheachothertofindthebestone.

Asforadaptivecontrol,useofameta-heuristicalgorithmhasbecomepopularrecently. Althoughthecommonfocusismotorcontrollerparametertuning,wenoticethepossi-bilities of applying meta-heuristic based adaptive control to our use cases.Furthermore, thegeneticalgorithmisselectedduetoitsthoroughresearchandabundantresources. Considering that during the construction of the chromosome, history power readings are required;amathematicalmodelcanrelievethesystemfrommaintainingalargehistory database. Thismodelcanbegeneratedusingregressionanalysistechniquessuchas polynomialregression.

3Methodology

Atthebeginningofthischapter,ageneraloverviewofthedevelopmentprocessare provided,whichwillbelaterusedintheimplementations.Thesubsequentpartswillbe ingreaterdetails.Firstofall,thegenerationofdataaswellassimulationoftheservo motor are elaborated;then the implementation of Predictive Maintenance and Adaptive Controlarepresented;finally,thecloudAPIforIoTbasedKnowledgeManagementare introducedbriefly.

3.1OverallFramework

ThisthesisworkisapartoftheCHARIOT[8]project.Ingeneral,CHARIOTisan IoTmiddleware,whichfocusesontheknowledgemanagementamongIoTentities.It employs an Industry 4.0 use case that includes the simulation of a full-scale autonomous smartfactoryenvironment.Figure3.1presentstheIndustry4.0smartfactoryusecase ofCHARIOTproject.

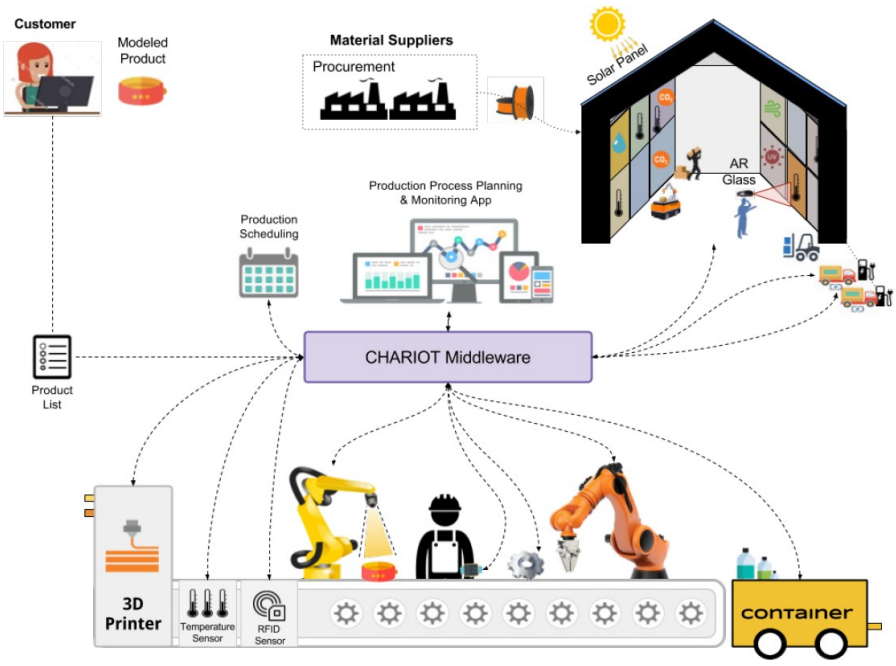


Figure3.1:TheusecaseoftheCHARIOTproject

The work done in this thesis mainly includes data generation and simulation;implementationofthepredictivemaintenancesemi-supervisedanomalydetector;implementation of the regression based adaptive control mechanism and the implementation of the Cloud APIforIoTbasedknowledgemanagement.

InFigure3.2,aflowchartisgiventobetterdescribethegeneraldevelopmentpro-cessofthesystem.

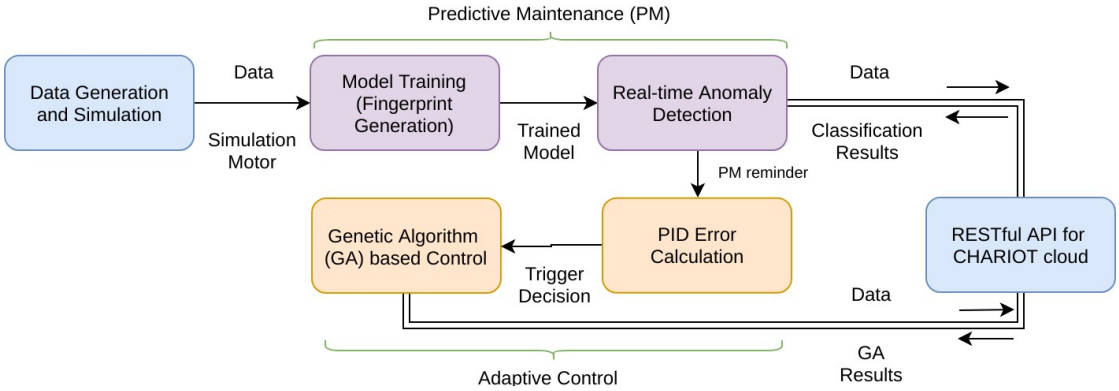


Figure3.2:Thesystemdevelopmentflowgraph

The first component in the flow chart is responsible for generating normal running data, includingspeed,powerandtorquefromtheservomotor.Thesedatawillbefurther processedandlaterbeusedinfingerprintgenerationforthepredictivemaintenance. Moreover,thedatawillalsobeusedtoconstructasimulationmodelfortheservoin Matlab.

Predictivemaintenanceisconductedinasemi-supervisedway,thereforeconsistsoftwo components.Thegoalofthesecondcomponentisusingclusteringalgorithmstotrain modelsfortheservomotor.Ittakesspeedandpowerfrommotor'snormaloperation dataasinputsandtrainsamodelfortheanomalydetection,whichisalsocalledfinger-printgeneration.Furthermore,thepurposeofthethirdcomponentistousethetrained modelinthelastcomponenttoperformpredictionsfornewincomingdata.Themodel trainingandpredictionfornewdataaredecoupled,sothatthetrainingcanbedone beforehandratherthanduringrun-timetosavecomputationtime.

Adaptivecontrolcontainstwocomponents.ThePIDerrorcalculationmoduleisre-sponsibleforcallingthegeneticalgorithmwhenthecalculatedPIDerrorisbiggerthan thethreshold.Then,theGeneticAlgorithmbaseddecisionmakingisusedtoselectthe bestspeedintervalsunderspecificconditions/requirements.

Lastbutnoleast,afterfinishingtheimplementationofpredictivemaintenanceand

adaptivecontrolmechanisms,aRESTfulAPIforCHARIOTcloudisdeveloped.This APIisresponsiblefor:(1)anydataexchangebetweenoursystemandtheMySQL database; (2)anyoperationthatisrelatedtomachinelearningorgeneticalgorithms.

3.2DataGenerationandSimulation

3.2.1DataGeneration

As described in the last section, data generation, more specifically, normal motor running datageneration,wouldbethefirsttaskofthisthesiswork.Onlywiththegenerated normalrunningdata,wecanproceedtothenextstages,i.e.simulationmotormodel constructionormodeltrainingforpredictivemaintenance.

3.2.1.1NormalData

Inordertogeneratenormaloperationdataformodeltraining,aRobotOperatingSys-tem (ROS) service called **generateTrainingData** is implemented.In this service, users canspecifythestartingandendingspeed,thenumberofstepsandthesimulationtime foreachstep.ThetrainingdataarecollectedusingtherosbagtooltolistentotheROS topic**motordataoperated**,whichshowsallthedataweneed,suchastimstamp, velocity,torque,current,effectivevoltage,power,etc.ThecodesnippetinListing3.1 showsastandardmotornormalrunningdatagenerationservicecall.Thespeedin-creasesfrom0to35r/min,withastepof0.5r/min;ateachspeed,itmaintainsfor 6secondsfordatacollection.Inpractice,althoughtheresolutionofthemotorspeed controlis0.229r/min,consideringthefrictionimbalanceontheconveyorbelt,itmakes nosensetouseasmallerspeed.Moreover,withasteptimeof6s,themotorwillhave enough time to reach a stable target speed.In the post-processing stage, the data where themotorspeedfluctuatestoreachthetargetspeedwillbeexcluded.

```
rosservicecall/conveyor_control_app/generateTrainingData"{index:1,
    ,→starting_speed:0.0,ending_speed:35.0,numberofsteps:70.0, ,→step_time:6.0}"
```

Listing3.1:TheROSservicefortrainingdatageneration

3.2.1.2AbnormalData

To test the effectiveness of the predictive maintenance and adaptive control mechanisms, abnormaldataarealsorequiredintheexperiments.Forrealmotor,thesedatacanbe generatedbyapplyingextratorqueontheconveyorbelt,forexample,addinganextra frictionpadtotheconveyorbelt,usinghandstopresstheconveyorbelttoincreasethe frictioncoefficient,etc.

However,inthesimulationmodel,suchextrafrictionneedstobesimulatedthrough functions.Therefore,adedicatedROSservice**generateAbnormalTorque**isimple-mented to simulate the gradually increasing abnormal torque as indicated in Listing 3.2.

```
rosservicecall/conveyor_control_app/generateAbnormalBehavior"{
,→isEnabled:true,index:1,type:'step',param1:0.2,param2:0.0, ,→holdTime:0.0}"
```

Listing3.2:TheROSserviceforgeneratingabnormaltorque

Inthisservice,fivedifferentfunctionsaresupported,includingstep,ramp,linear,log-arithm,exponentialandsinusoidalfunctions.Userscanspecifytwoparametersandan additional "holdtime" to control the behaviour of the generated abnormal torque.When serviceiscalled,astartingtimeisset,theservicecalculatestheabnormaltorquebased on$T=f(\Delta t)$,whereTisthecalculatedtorque,Δtisthetimedifferenceand$f$isthe pre-definedfunction.Specificdefinitionsofeachfunctionareshownasfollows:

•logarithm:

$$T= log(a \cdot \Delta t + b) \tag{3.1}$$

•exponential:

$$T= exp(a \cdot \Delta t) + b \tag{3.2}$$

•linear:

$$T= a \cdot \Delta t + b \tag{3.3}$$

•ramp:

$$T= a \cdot \Delta t \tag{3.4}$$

•step:

$$T= a \tag{3.5}$$

•sinusoidal:

$$T=a\ 2\cdot sin(2\pi b\cdot \Delta t +3\pi 2) \tag{3.6}$$

Where$a$and$b$aretwoparameters.Additionally,ifthe"holdtime"valueisgiven,the abnormal torquewill remain unchanged after "holdtime".Anexplanatory figure forthe sinusoidalfunctionisshowninFigure3.3.

3.2.2Simulation

Inthisthesiswork,asimulationmodelfortheservomodelisalsobuiltasthe"digital replica"oftherealmotor,sothatwecantesttheeffectivenessofouralgorithmsbefore applyingtotherealmotor.Insomecaseswheretherealmotorisstronglyaffectedby thenoises(e.g.imbalanceoffrictionontheconveyorbelt,inaccuracyofthecurrent sensor,etc.),thesimulationmotorisusedtobetterevaluatetheinfluenceofdifferent parametersonthealgorithmperformance.

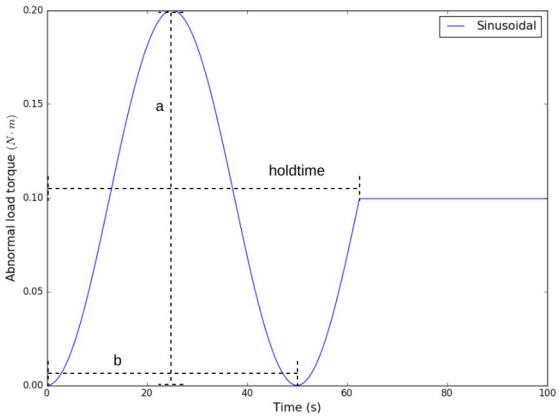


Figure3.3:Thesinusoidalfunction

3.2.2.1Implementation

ThesimulationisbasedonaDCmotorcontrolexamplecalled**pedcmotorcontrol** inMatlabSimulink.Forsimplicityandperformancereasons,themodelislargelymod-ified.Forexample,tosimplifythemodel,theoriginaldoubleloopPIcontrolandthe pwm generator are deleted.Instead, a PID controller and a controlled voltage source are addedtodirectlycontroltheinputvoltagetothemotor.Animplementationoverview ofthesimulationmotorisshowninFigure3.4.

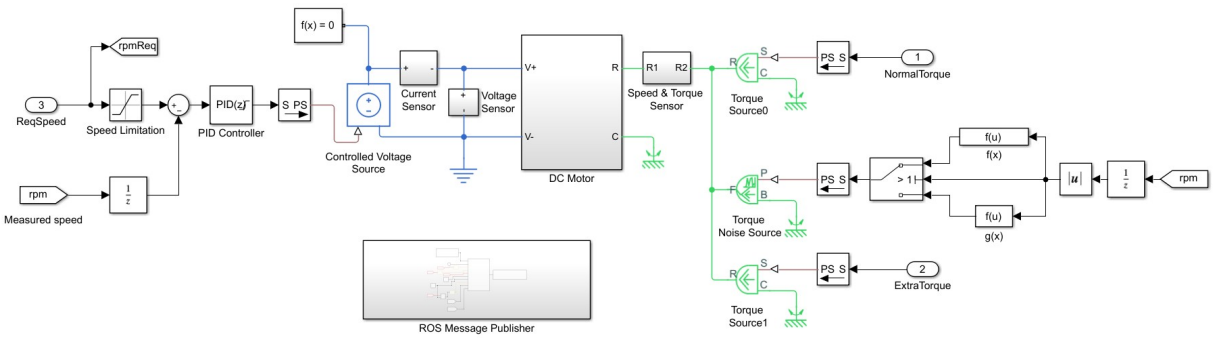


Figure3.4:Implementationoverviewofthesimulationmotor

Moreover,tomakethesimulationmodelcompatiblewiththeROSenvironment,addi-tional modifications needed to be made.First of all, the simulation motor should be able to receive data from ROS services, topics or something equivalent, so that one can easily controlthesimulationinROSenvironment.ConsideringthatSimulinkdoesnothave verygoodsupportforROSservices,weusedROStopicsandROSparameterstopass information.Basedonourusecases,wedecidethatthereshouldbefourparameters thatarecontrolledfromoutside:aswitchtoturnon/offthemotor(**isEnabled**),the

motorspeed(**speed**),thenormalloadtorque(**torque**)andtheabnormalloadtorque (**extratorque**).Theseparationofnormaltorqueandabnormaltorquemakesitpossibletosimulateaconstantloadtorqueasthebasis,i.e.”normalsituation”.Then, any deviation from the basis would be considered as abnormal.Since the **extratorque** wasassumedtobechangingallthetime,itwasimplementedasaROStopic;while theotherswereallimplementedasROSparameters.Figure3.5showstheparameter receivingblocksinthesimulation.

Moreover,thesimulationmotorshouldbeabletosenddatatotheothercomponents inthesystem.Similartotherealmotor,theoperationdataoftheservomotorispublishedtoatopiccalled**motoroperateddata**.Aseparateblockisimplementedfor thepublishingasinFigure3.6.

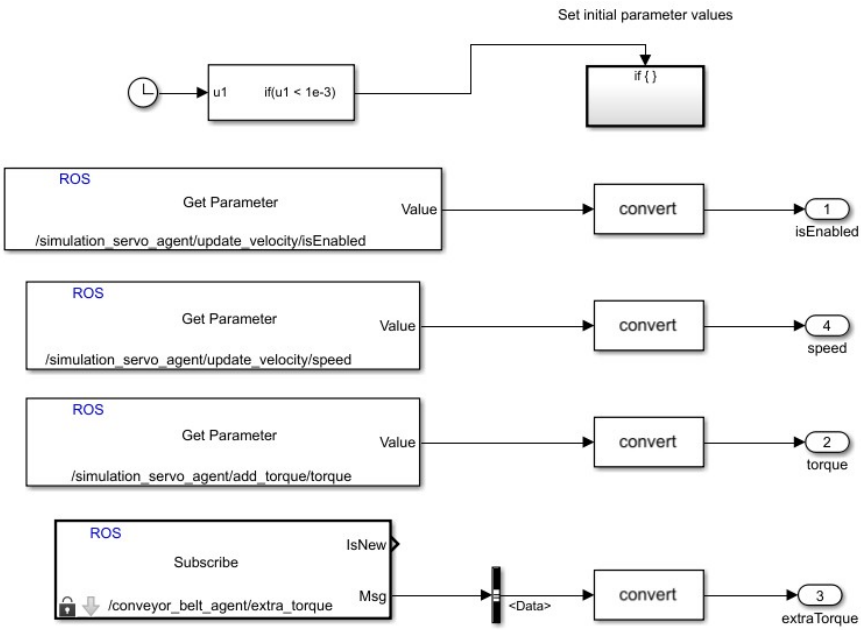


Figure3.5:TheROStopicandROSparameterreceivers

Additionally, to make the behaviours of the simulation motor to be as close as possible to therealmotor,onealsoneedstoconsiderabouttheenoisescausedbyfrictionimbalance ontheconveyorbeltandtheinaccuracyofthecurrentsensor.Therefore,firstofall, aGaussiannoiseisaddedtotheloadtorqueofthemotor,whichwillbecomeslightly largerwhenthespeedishigher;similarly,anGaussiannoiseincreasingwiththespeed isaddedtothecurrentsensorreading;lastbutnotleast,themotorspeedreadingsare roundeduptoreflectthespeedsensorresolution0.229r/minontherealmotor.The implementationofthesemeasuresisshowninFigure3.7.

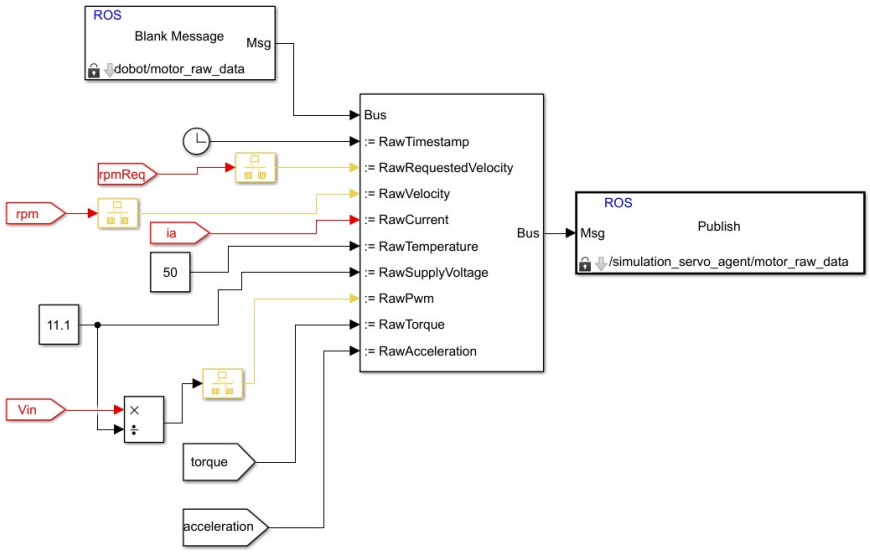


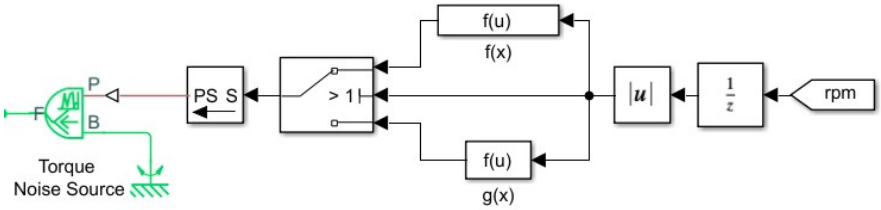
Figure3.6:TheROStopicpublisher

3.2.2.2ConfigurationoftheSimulationVariables

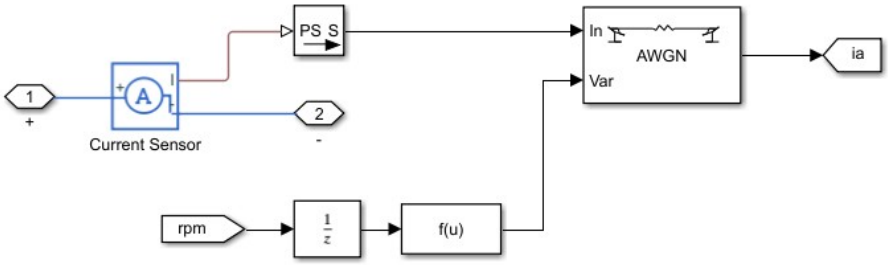
Inthissimulationservomotor,therearesevеralparametersthatdefinethecharacter-isticsofthesystemthatneedtobeconfigurеdduringthemodelconstruction.These parametersareinertia(J),inductance(L),resistance(R),motorbackemf(K),viscous frictioncoefficient(F)andbreakawayfrictiontorque(C).CandFdefinethesimplified model of the friction inside the servo motor system, i.e.$T= C+F \cdot\omega$ (where ω is motor's angularspeed),whichaffecttheslopeandinterceptofthespeed-currentbehaviour.K definestheelectricpower-kineticpowerconversionrate,i.e.$V= K \cdot \omega$,whereVisthe inputvoltageandωistheangularspeedofthemotor.WhileRdefinestheresistance inside the motor.K and R together affect the slope and intercept of the speed - effective voltagebehaviour.Additionally,inertiaJandinductanceLaffecttheresponsetimeof thesystem.

Frompreliminarytestresultswelearnedthat,specifically,Cisrelatedtotheintercept ofthespeed-currentbehaviour,whileFisrelatedtotheslope.Likewise,parameterK isrelatedtotheslopeofthespeed-effectivebehaviour,whileRisrelatedtotheinter-cept.TestresultsalsoindicatedthatchangingthevaluesofKandRwillalsoaffectthe current,whileCandFwillnotaffectthevoltage.ThissuggestedthatKandRshould be tuned first.Considering the close relations between K and R as well as C and F, they areseparatedintotwogroupsforparametertuning.Since$power=current *voltage$, itisclearthataslongastheeffectivevoltageandcurrentbehavioursofthesimulation motorareclosetothoseoftherealmotor,theirpowerbehaviourswillalsobesimilar.

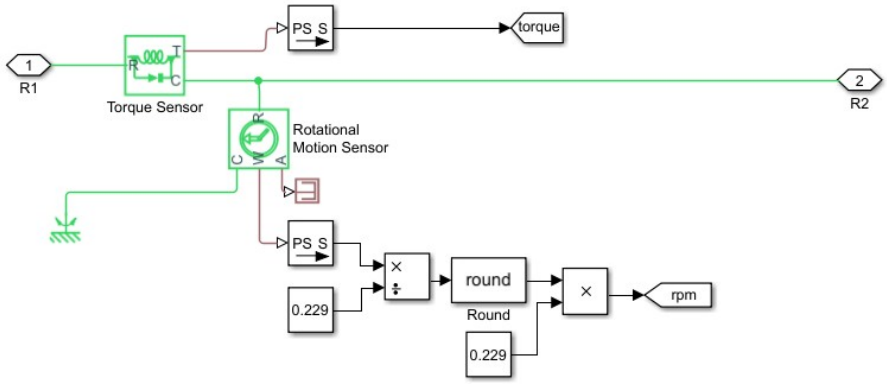
A simple yet effective twiddle program is developed to select the best parameters.It is a



(a)TheGaussiantorquenoisesource



(b)TheGaussiannoiseaddedtothecurrentsensor



(c)TheGaussiannoiseaddedtothespeedsensor

Figure3.7:The parameter receiving and publishing mechanisms in the simulation motor

trial and error algorithm which tries to approximate the desired target gradually.When itsensestheerrorbecomessmallerinonedirection,ittriestoincrease(1.1times)the trial step in that direction in the next iteration ;otherwise, it tries the other direction;if both directions don't work,the trial step will be shrunk (0.9 times),and the parameters willreturntotheoriginalvalueatthebeginningofthisiteration.Pseudocodeforthe twiddleprogramisshowninListing3.3.

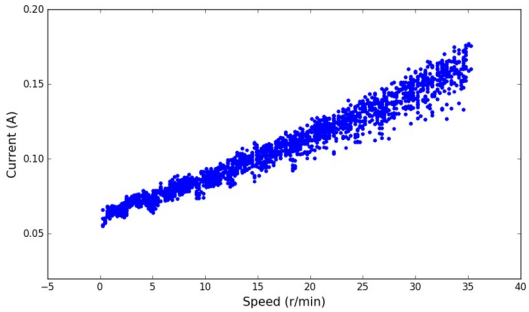
```
Input:DesiredmaximumerrorarrayRequestedError[2]
Ouput:CalculatedparameterarrayParams[2]
Variables:Err[2],Delta[2],BestErr[2],BestParams[2]

deftwiddle():
while(Err[1]>RequestedError[1]orErr[2]>RequestedError[2])          fori=1:2do:
        ifErr[i]<=RequestedError[i]:
        Donothing,gotonextparam

        else:
          Trybiggervalue,Params[i]=BestParams[i]+Delta[i]
          CalculateerrorforParams[i]

          ifErr[i]<BestError[i]:
            RecordErr[i]andParams[i]asBestError[i]andBestParams[i   ,→];
            Increasetrialstep,Delta[i]=Delta[i]*1.1;

          else:
            Trysmallervalue,Params(i)=Params(i)-2*Delta(i); CalculateerrorforParams[i];

            ifErr[i]<BestError[i]:
            RecordErr[i]andParams[i]asBestError[i]andBestParams[i   ,→];
            Increasetrialstep,Delta[i]=Delta[i]*1.1;
            else:
            Gobacktotheoriginalvalueinthisiteration,Params(i)=          ,→Params(i)+Delta(i);
            Shrinktrialstep,Delta(i)=Delta(i)*0.9;
```

Listing3.3:Pseudocodefortwiddleprogramtocalculatetheoptimalparameter combinations

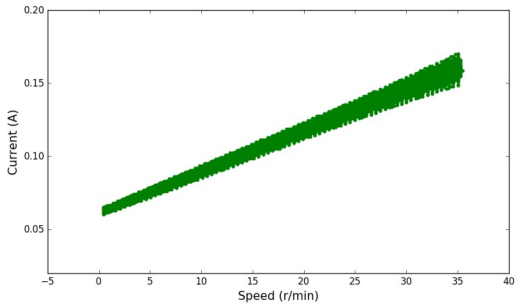
TheselectedparametersareshowninTable3.1.Thecorrespondingsystembehaviours oftheservomotorareindicatedinFigure3.8.

Variables	J	L	K	R	C	
Values	6e-4	3.5	6e-3	12	6.2636e-7	3.6

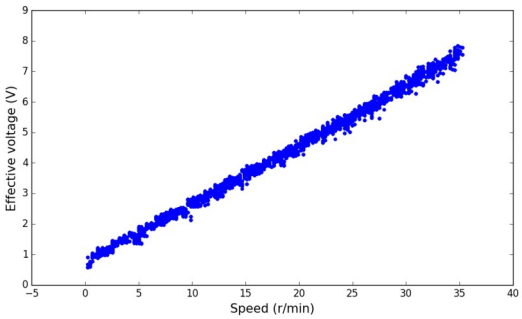
Table3.1:Thetunedvariablesforthesimulationmotor



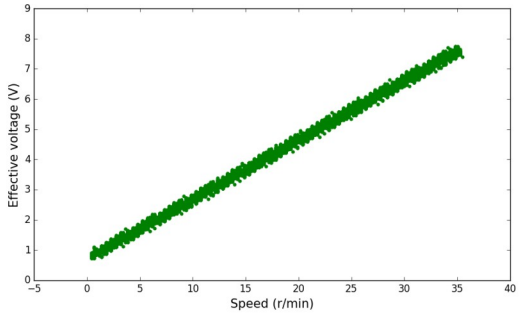
(a)Thecurrent-speedbehaviourofthereal



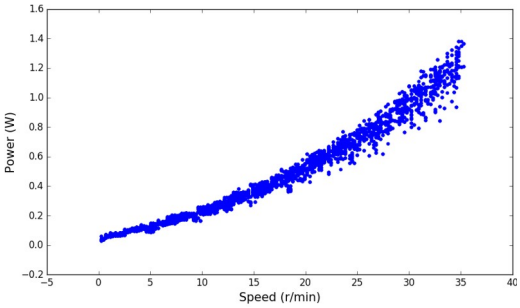
(b)Thecurrent-speedbehaviourofthesimulationmotor



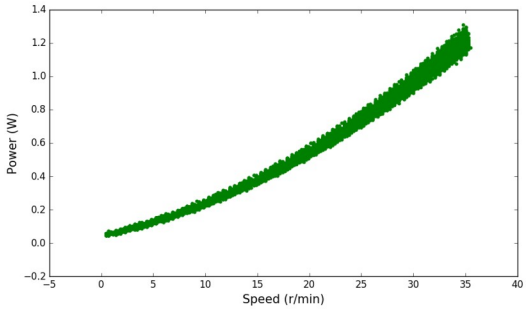
(c)Theeffectivevoltage-speedbehaviourofmotor



(d)Theeffectivevoltage-speedbehaviourof therealmotor



(e)Thepower-speedbehaviourofthereal thesimulationmotor



(f)Thepower-speedbehaviourofthesimu-
motor lationmotor

Figure3.8:Thecomparisonbetweentherealmotorandthesimulationmotor

3.3PredictiveMaintenance

Inrecentyears,noveltydetectionandanomalydetectionhavebeenwidelyappliedin financialfrauddetection,intrusiondetection,faultdetection,predictivemaintenance, etc.Duetotheirgoodperformanceandpromisingfuture,evenmoremethodsarere-discoveredandadjustedtobeusedinthisarea.Goodreviewsonmethodsinnovelty detectionandanomalydetectionareprovidedin[27],[32],[61]and[62].

Inthisthesiswork,threecommonlyusedmethods:oneclassSVM,*k*-NNandkernel PCAareimplementedandcomparedforourusecases.Thefollowingsectionsdescribe thesemethodsdetailedly.

3.3.1OneClassSVMbasedAnomalyDetection

One class support vector machine is a popular domain-based novelty detection algorithm. Originally,supportvectormachineisusedinclassifications.Itgeneratesanon-linear decision boundary among different classes by projecting the data to a higher dimensional space [63].Specifically, considering the following data set $\Omega = (x_1, y_1), (x_2, y_2), ..., (x_i, y_i)$ where $x_i \in \mathbb{R}^d$and $y_i \in \{-1, 1\}$ are the *i*th input data point that exists in the input space *P*and *i*th classification output, respectively.In most of the classification problems, data pointsarenotlinearlyseparable.Insupportvectormachine,thisproblemissolvedby projectingthedatapointsfrominputspace*P*tothefeaturespace*Q*,inwhichthedata points can be separated by a ”linear” hyperplane.Later on, by projecting the hyperplane back to the input space, one can acquire a non-linear curve that can distinguish the data points.Thehyperplaneinthefeaturespace*Q*canberepresentedas$w^T x + b = 0$, where*w*isthenormalvectortothehyperplane.Moreover,definethemaximaldistance betweentwoparallelhyperplanesthatcanseparatethe $\frac{2}{|w|}$,then the goal of the algorithm is to maximize.Furthermore,to avoid the over-fitting problem causedbyoutliers,usuallyaslackvariable$\xi_i$isadded,andtheconstant*C*decideshow bigthetrainingerroris,i.e.howmanytrainingdatapointsareallowedtobeinside themargin.Ingeneral,thealgorithmofsupportvectormachinecanbesummarized representedas:

$$\min_{w,b,\xi_i} \frac{\|w\|^2}{2} + C \sum_{i=1}^{n} \xi_i \tag{3.7}$$

subjectto:

$$y_i(w^T \varphi(x_i) + b) \geq 1 - \xi_i, \xi_i \geq 0 \tag{3.8}$$

Anotherimportantconceptinsupportvectormachineisthekernel”trick”.Thekernel function $K(x, x') = \varphi(x)^T \varphi(x_i)$ provides a shortcut such that as long as *K*has the same resultsastheoriginalfunction,itcanbeusedinstead.Thekerneltrickhaslargely

increased the usability of the algorithm.The most widely used kernel is the Radial basis functionkernel(RBFkernel),whosekernelfunctionis:

$$K(x, x') = exp\left(-\frac{\|x - x'\|^2}{2\sigma^2}\right) \quad (3.$$

FirstproposedbySch¨olkopfetal.[31],oneclassSVMtransferstheideaoftheoriginal supportvectormachinetothedomainofunsupervisedlearning.Unlikeusualunsupervisedlearningalgorithms,theclusteringresultwouldbeonlyoneclass.Itconstructs abinaryfunctionthatcapturestheregionswheretheprobabilitydensitylives[64].In other words in the region where most of the data points distribute, the function will give thevalue+1,and-1elsewhere.Themechanismbehindoneclass SVMis,inthefirst place,mappingthedatapointsintothefeaturespacethatcorrespondstothechosen kerneljustliketheoriginalSVMalgorithm.Then,insteadofmaximizingthemargin betweentwogroupsofdatapoints,thealgorithmtriestomaximizethemarginbetween thedatapointsandtheoriginusingahyperplane[32].Thefunctionbecomesslightly differentfromtheoriginalone,butonecanstillspottheresemblance:

$$\min_{w,b,\xi_i} \frac{\|w\|^2}{2} + \frac{1}{\nu l}\sum_{i=1}^{n} \xi_i - \rho \quad (3.$$

subjectto:

$$(w^T\varphi(x_i)) \geq \rho - \xi_i, \xi_i \geq 0 \quad (3.11)$$

Whereνisanimportantparameterthat:(1)definestheupperboundofthefractionof outliers; (2)definesalowerboundofthenumberofsamplesthatareusedassupport vector.

3.3.1.1ImplementationandParameterSelection

In this work, the one class SVM class **svm.OneClassSVM** from scikit-learn [21] is used. It provides the full functionality of the one class SVM algorithm, while well encapsulated foreasyuse.Threefunctions,namelyfit(),predict()anddecisionfunction()areused totraintheoneclassSVMmodel,performclassificationonsamplesandacquirethe decisionboundaryofthemodel.

InoneclassSVM,theparametersν(hyperparameteroftheoneclassSVMalgorithm) andγ(hyperyparameteroftheRBFkernel)willlargelyaffecttheperformanceofthe algorithm. [2]and[65]suggestthatνcontrolstheupperboundofthetargetdatathat areexcluded,whichisusuallytunedtorejectnoise.Whileγcontrolsthesmoothnessof thedecisionboundary;thebigger$\gamma$is,themorejaggedtheboundaryis.

Normally,onecanperformamodelselectiontoselectthebestparameters;however, inthecaseofoneclassSVM,thelackingof”abnormaldata”makesitimpossible.

Atypicalwaytosolvethisproblemistogenerateartificialoutlierdata[32]. Liet al.inpaper[66]proposed$k$-NNbasedEdgePatternDetection(EPD)methodto identifyseveraldatapointsthatdefinetheedgepatternofthetrainedoneclassSVM model.ThemechanismbehindEPDisinstinctive:foradatapointinthetrainingset, $v_{ij}=\frac{x_i-x_{ij}}{\|x_i-x_{ij}\|}$, j=1, 2, ..., $k$representstheunitdirectionvectorfromits$j$thk-nearest-neighborx_{ij}toitself.Furthermore,thenormalvectorofthetangentplaneat$x_i$can beapproximatedby$n_i=\sum_{j=1}^{k} v_{ij}$.Thewayofdecidinganedgepatternistocheck whether for this point,all or most v_{ij}satisfy $v_{T\,ij}\cdot n_i\geq 0$.An explanatory graph for EPD algorithmisshowninFigure3.9.

Furthermore,Wangetal.inpaper[2]proposed”negativeshifting”togeneratehigh quality outlier based on the previous EPD algorithm as in Figure 3.11.Intuitively,good outlierscanbegeneratedbyshiftingtheidentifiededgepatternawayfromthetarget databyashortdistance.Toachievethat,Wangsuggeststheshiftingdirectiontobe targetdatadensity’snegativegradient,wherethedensityoftargetdatadropsatthe fastestrate,whichcanbedirectlyestimatedusingthe$n_i$weobtainedfromtheedge patterndetection:

$$-\nabla p(x) \sim \mathbf{n} = \sum_{j=1}^{k} \frac{x_i - x_{ij}}{\|x_i - x_{ij}\|} \tag{3.12}$$

Moreover,thedistancel_{ns}betweenthepseudooutlierandtheedgepatternshouldnot biggerthanthefurthestk-nearestneighboroftheedgepattern.Therefore,itissafe totaketheaveragek-nearest-neighbordistanceofalltheedgepatternsasourshifting distancel_{ns}.Ingeneral,thepseudooutliercanbecalculatedas:

$$l_{ns} = \frac{1}{|X_{edge}|} \sum_{x_i \in X_{edge}} \frac{1}{k} \sum_{j=1}^{k} \|x_i - x_{ij}\| \tag{3.13}$$

$$X_{outlier} = \{ x_{(i)\,o} | x_{(i)\,o} = x_i + \frac{n_i}{\|n_i\|} \tag{3.14}$$

Additionally,Wangproposed”positiveshifting”togeneratepseudotargetdataasa validation set to avoid time-consuming cross validation as in Figure 3.10.The mechanism behind”positiveshifting”isactuallysimilarto”negativeshifting”:thepseudotarget datacanbegeneratedbyshiftingeverytargetdatapointalongthedirectionwhere targetdatadensitygrowsatitsfastestrate,whichisexactlytheoppositedirectionof the negative shifting.Moreover, the shifting distance should be the minimum projection distance of this data point’s k-nearest neighbors on the shifting direction.Therefore,for any target data point $x_i$and its k-nearest neighbors x_{ij}, j= 1, 2, ..., k, the corresponding pseudotargetdatapointcanbecalculatedas:

$$X'_{target} = \{ x_{(i)\,t} | x_{(i)\,t} = x_i + \frac{\nabla p(x)}{\|\nabla p(x)\|} \cdot \left(\min_{x_{ij}} \frac{\nabla p(x)}{\|\nabla p(x)\|} \cdot (x_{ij} - x_i) \right), \forall x_i \in X_{target} \} \tag{3.15}$$

Where $\nabla p(x)$ isthepositivedirectionoftargetdatagradient,whichhasacontrarysign ofEquation3.12.$x_{min+\ ij}$ representstheneighborthathastheminimumprojectiondistanceonthepositivedirectionoftargetdatagradient.

Applyingthese"negativeshifting"andpositiveshiftingtoourtrainingdataset,we cannowobtainartificialoutliersandpseudotargetdata.Theerrorofthesystemis calculatedas:

$$Err = 0.5 \cdot Err_o + 0.5 \cdot Err_t \tag{3.16}$$

Details of the artificial outliers and pseudo target data generation results and parameter selectionresultsareshowninSection4.2.1.1.

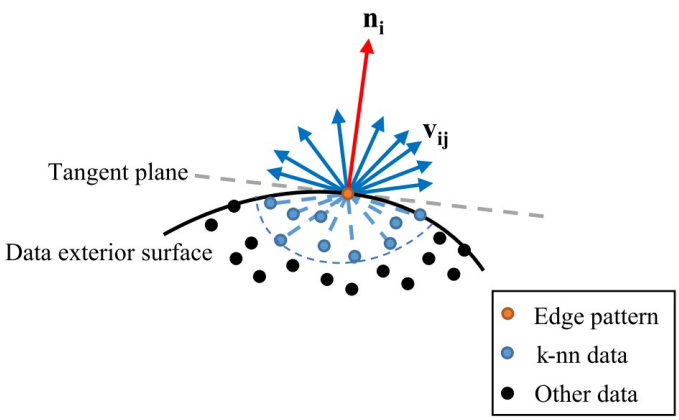


Figure3.9:TheEdgePatternDetectionalgorithm[2]

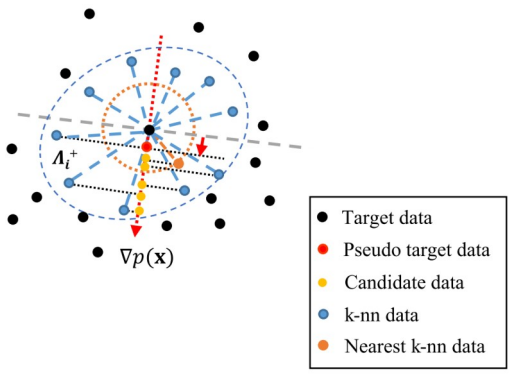


Figure3.10:Thepositiveshiftingalgorithm[2]

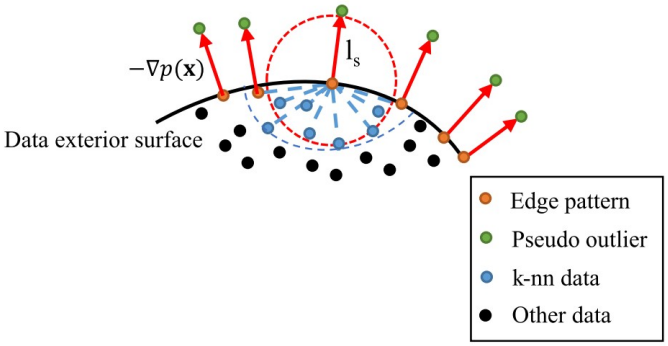


Figure3.11:Thenegativeshiftingalgorithm[2]

3.3.2k-NNbasedAnomalyDetection

The*k*-NNapproachisawidely-useddistancebasednoveltydetectionmethod.Itis astraightforwardapproachbasedontheassumptionthatnormaldatapointsallhave closeneighbors.Thereforeifadatapointisfarawayfromitsneighbors,itisregarded asanoutlier[28].Despiteitssimplicity,beinganon-parametricmethod,itisoftensuc-cessfulinclassificationsituationswherethedecisionboundaryisveryirregular,hence verysuitableforourusecasehere.

Generally,the*k*-NNbasedanomalydetectionapproachisworkinglikethis:firstofall, for every data point in the training dataset,the k-nearest neighbors of the point have to befound.Then,togetthenoveltyscore,severalapproachescanbechosenfrom:either thedistancetothekthnearestneighborisused[29],ortheaveragedistancetoallthe k-nearestneighborsiscalculated[27],orthemediandistancetothek-nearestneighbor iscomputed[30].Itisnotablethatthecalculatednoveltyscoreaboveisonlyforone singledatapoint;tocalculatethenoveltythresholdforthewholetrainingdataset,the algorithmneedstocalculatethenoveltyscoresforallthedatapoints,andchoosethe biggestamongthemtobethethreshold. Finally,topredictwhetheranewcoming datapointisanovelty,thealgorithminsertsthepointintothetrainingdataset,finds itsk-nearest-neighbors,calculatesitsnoveltyscoreandcomparesthescorewiththe threshold.

3.3.2.1ImplementationandParameterSelection

In this work, the class neighbors.NearestNeighbors from Scikit-learn is used.Key param-etersfor*k*-NNincludethenumberofthenearestneighbor*k*andtheleafsize*leafsize* ofthecalculatedBDtreeorBallTree.Accordingto[67]and[68],thebigger*k*is,the smootherthedecisionboundaryis,thereforeitcanalsobeusedtosuppressnoise; however,theselectionofkvaluelargelydependsonthenatureoftheproblem.While *leafsize* defines the minimum cell size in KDTree or BallTree; smaller leaf size indicates

amorecomplicatedtree.Accordingto[68],larger*leafsize*leadstoafastertreecon-struction time and less consumed memory space.As for query time, the closer *leafsize* isto1,thehigherquerytime;thecloser*leafsize*istothetrainingsetsize,theworse performance. [68]suggestsagoodbalanceis*leafsize* = 30,whichisthedefaultvalue. Theparameterselectionresultsof*k*-NNispresentedinSection4.2.1.1.

3.3.3KernelPCAbasedAnomalyDetection

PCAisareconstructionbasedapproach.Itusesorthogonaltransformationtoreduce thedimensionsofthedatawhilemaintainingitsoriginalinformationasmuchaspossi-ble.Thereforeitiswidelyusedinreducingthedimensionsofthedata.Inrecentyears, researchersdiscoveredthatitalsoprovidesanewviewpointtodistinguishnoveltyin thetransformedspace.HeikoHoffmann[36]proposedamethodtoapplykernelPCA tonoveltydetection. Accordingtohim,thisapproachhasshowngoodperformance ifcomparedtooneclassSVMandparzenwindowdensityestimator.Themechanism behindkernelPCAis,bytransformingthedatafromtheoriginalfeaturespacetothe infinite-dimensionalspaceandviceversa,thereconstructionerrorgeneratedinthepro-cesscanbeviewedasanoveltymeasure.ItcanbecalculatedbytheEuclideandistance betweentheoriginalmatrixandthereconstructedmatrix:

$$Err = \|X - X_{re}\|_2 = \sum_{i=1}^{m} \|x_i - \hat{x}_i\|_2 \tag{3.1}$$

3.3.3.1ImplementationandParameterSelection

Inthiswork,thekernelPCAanomalydetectionapproachisimplementedusingclass decomposition.KernelPCAfromthescikit-learn[21]library,basedonthepaperfrom HeikoHoffmann[36].Morespecifically,fit(),transform()andinversetransform()func-tions are used to train the model, transform the input data to infinite-dimensional space andinversetransformthedatabacktooriginalspace,respectively.Thereconstruction erroristhencalculatedbyL2norm,usingfunctionnumpy.linalg.norm().Additionally, theinfluencesofthehyperparameterγinRBFkernelandthehyperparameterαofthe ridgeregressionthatlearnstheinversetransformontheperformanceofthealgorithm are also tested and compared.According to [69] and [70], *alpha* stands for regularization strength in ridge regression; the larger α is, the less variance the estimates have.In other words, decreasing the value of α will lead to higher model complexity.While γserves the same purpose as explained in one class SVM in Section 3.3.1.1.The parameter selection resultsareshowninSection4.2.1.1.

3.3.4Comparison

Inthissubsection,threealgorithmsarecomparedtofigureoutwhichisthebestalgo-rithmintermsoftrainingspeed,predictionspeed,predictionaccuracy,etc.

The experiment results show that one class has the best performance in terms of predic-tion speed and prediction accuracy.Detailed experiments results and analysis are shown inSection4.2.1.2.Ingeneral,itshouldbeselectedasthebestcandidatetoimplement onthefinalalgorithm.

3.4AdaptiveControl

Asdescribedinthepreviousssection,predictivemaintenanceprovidesafastandsen-sitivemethodtodetectthepotentialfaultsinthesystemorthefluctuationsofload torque.However,beingabletodetectanomaliesisnotenough.Itwouldmakemore sensetochangethespeedofthemotoraccordingly,tominimizefurtherdamagetothe motor,ortosaveenergywhilemaintainingreasonableefficiency.Tobeabletodothat, traditionally,oneneedstorelyonarule-basedspeedcontrolapproach.Commonrules includesettinganupperboundpowerorcurrentlimitforthesystem.Nevertheless, arulebasedapproachisnotsatisfyinginbothofthecasesmentionedabove.Forthe system fault situation, since the threshold is fixed, the damages might have already been done before the system stops or slows down the conveyor belt.While for the load torque fluctuationsituation,assumingtheloadtorqueischangingallthetime,therule-based approachwillslowtheconveyorbeltdownwhentheloadtorquereachesthethreshold; butitwillnotspeedtheconveyorbeltupagainiftheloadtorquedropslater.This willcauseunnecessarylossofefficiency.Therefore,analgorithmwhichcanreacttothe changesoftheloadtorqueaccordinglyisdefinitelypreferredinboththesituations.In thissection,weintendtoimplementsuchanalgorithm.

3.4.1PolynomialRegression

To avoid maintaining a large history database for speed-torque-power readings as in mo-torfreepredictivecontrol,wedecidedtogoforamorestraightforwardway,i.e.model basedcontrolasdescribedinSection2.2.3.Sincewearenottryingtobuildamath-ematicalmodelforthemotor,itispossibletodirectlyuseregressiontocalculatethe relationsbetweenspeed,powerandtorque.

ApolynomialregressionisperformedusingMatlab’sCurveFittingToolboxbasedon the data we collected from the simulation motor results.As long as the simulation motor model is constructed accurately enough under no-torque condition , it is possible to sim-ulate the motor behaviours under different load torque using the same model.Including torqueasaparameterintheregression,theexpectedcalculationformulaisasfollows:

$$P = f(v, T) \tag{3.18}$$

WhereP,v,Tarethepower,velocityandtorque,respectively.

Inordertogenerateagoodmathematicalmodelusingpolynomialregression,wefirst runmotorsimulationsunderdifferentspeedanddifferenttorquetofigureoutwhatis the maximum torque the motor can withstand at a certain speed without slowing down. Then, the recorded maximum torque data are used as an upper bound to limit the speed in the simulation.The simulations were conducted in a nested loop manner:in the outer loop,thetorqueiteratesfrom0untilthemaximumtorquelimit,oruntilalimitsetby theusers;ineachiteration,thetorqueincrementsitselfby0.005$N \cdot m$.Intheinner loop,thespeediteratesfrom0tillthemaximumspeedthemotorcanmaintainunder thecurrentloadtorque;ineachiteration,thespeedincreasesby0.5r/min,andthe systemwillmaintainthecurrentspeedfor6s.Furthermore,tosavesimulationtime, insteadofdoingaexhaustionsearchforallthetorque/speedcombinations,wesetthe simulationspeedrangeas[0,40]andtorquerangeas[0,0.3].Thisshouldbesufficient, consideringthemostfrequentlyusedspeedrangeis0-35r/minandtheloadtorqueis usuallylightweightboxesinourexperimentsetups.InListing3.4,thepseudocodefor polynomialregressiondatagenerationispresented.

```
Input:speedrange[va,vb],torquerange[Ta,Tb],simulationtime ,→ ts,
speedsteplengthvi,torquesteplengthTi;
Output:2DpowerarrayP[];
Variable:currentpowerP;

defpolyDataGen():
    ts=6s
    va=0,vb=40,vi=0.5
    Ta=0,Tb=0.3,Ti=0.005

    foriin[Ta,Ta+Ti,Ta+2*Ti,...,Tb]do:
    forjin[va,va+vi,va+2*vi,...,vb]do:      P=simulation(i,j)
        waitts
        P[i,j]=P

    returnP[]
```

Listing3.4:Pseudocodefordatagenerationforpolynomialregression

Using the collected data,we set the degree for $v$and Tboth to 2,so that the model gets agoodperformancewhilenotgettingveryhighcomplexity.Thefollowingregression modelisabletobeconstructed:

$$P = 0.04276 + 0.01438 \cdot v + 0.9586 \cdot T + 0.0005536 \cdot v^2 + 0.1481 \cdot v \cdot T + 4.97 \cdot T^2 \quad (3.19)$$

Where the model receive an R-square (the closer to 1, the better) of 0.9987, which means the model is representing 99.87% of the variation of the data;and a Root Mean Squared

Error (the closer to 0, the better) of 0.02952, indicating the standard error of the model is low.Thesurface(representingthemodel)generatedbypolynomialregressionisshown inFigure3.12.Thismodelcanalsobefurtherappliedtoournotorquesituationby settingTintheregressionmodeltobe0.Thecorrespondingmathematicalmodelis:

$$P = 0.04276 + 0.01438 \cdot v + 0.0005536 \cdot v^2 \quad (3.20)$$

Whiledirectlyperformpolynomialregressiontomodeltherelationsbetweenpowerand speed,wehave:

$$P = 0.04421 + 0.01424 \cdot v + 0.0005544 \cdot v^2 \quad (3.21)$$

Equationsindicatethatthereduceddimension3Dmodel(P=f(v,T)whereT=0)has comparableperformancetothedirectlygenerated2Dmodel(P=f(v)).Furthermore, theresemblanceoftwomodelsarealsoshowninFigure3.13.

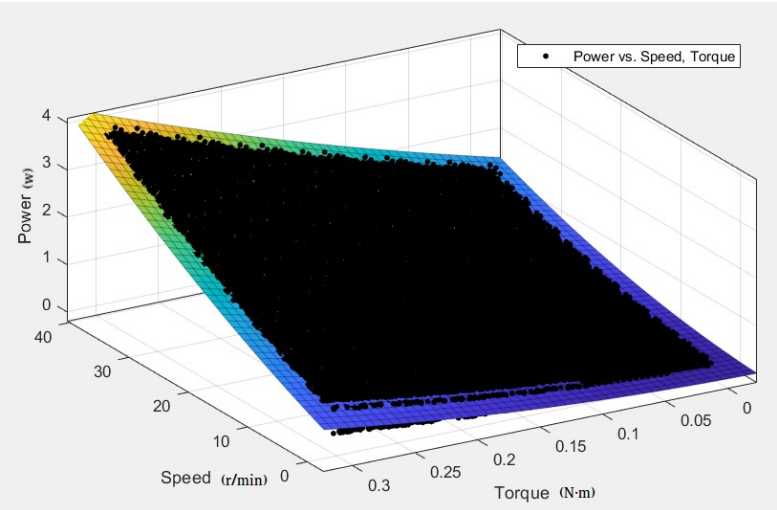


Figure3.12:Thesurfacegeneratedbypolynomialregression

3.4.2GeneticAlgorithmbasedControl

3.4.2.1Requirements

Ourtargetistoimplementanalgorithmthatcanfulfillthefollowingrequirements:

- The proposed algorithm should be able to adjust the motor speed according to the currentloadtorque.Iftheloadtorquegoesup,themotorshouldslowdown;if theloadtorquegoesdown,themotorshouldspeedup.
- Theusershouldbeabletoselectwhetherpowersavingortime-savingispreferred byspecifyingcontrolparameters.

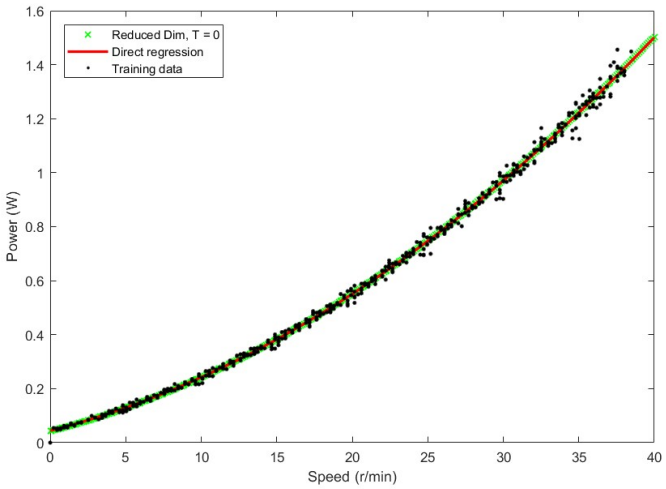


Figure3.13:The comparison between the reduced dimension 3D model (P=f(v,T) where T=0)andthedirectlygenerated2Dmodel(P=f(v))

3.4.2.2ConstructionoftheChromosomeSpace

Thechromosomespacecontainsfourgenes,whicharelowerboundspeed,upperbound speed,lowerboundpowerandupperboundpowerasindicatedinTable3.2.Thedis-tancebetweenthelowerandupperboundofthespeed,i.e.thespeedinterval,canbe changedwhencallingtheservice;however,duringeachservicecall,thespeedinterval staysthesame.Anexplanatorygraphonhowtoacquiretheupperboundandlower boundofthepowerispresentedinFigure3.14.

No.	1	2	3	4
Name	speedmin	speedmax	powermin	powerma

Table3.2:Thelayoutofthechromosomespace

Thankstotheregressionmodelwebuiltfromthelastsubsection,wecancalculate thecorrespondinglowerandupperboundsofpowerusingtheregressionmodel,instead of getting the values through experiments.The chromosome space is constructed in this waysothatwecaneasilycompareandtesttheperformanceofdifferentspeedintervals.

3.4.2.3FitnessFunction

ThefitnessfunctioninthetraditionalGeneticAlgorithmuseshistoricaldatatomake decisions.However,in our case,**it is not realistic to noly use historical data to deal with**

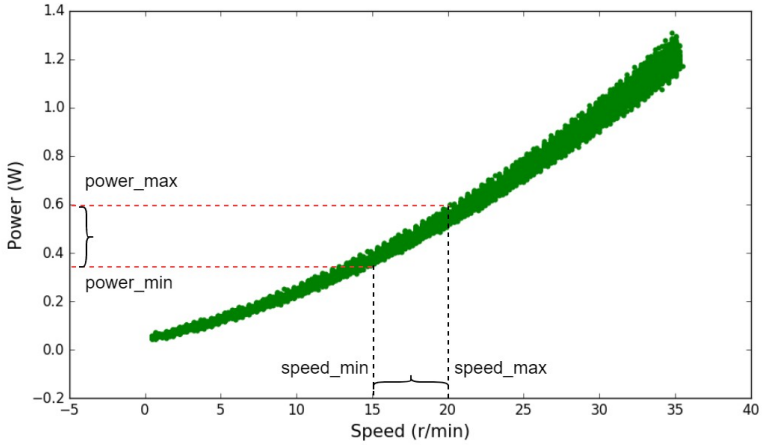


Figure3.14:Theupperandlowerboundofpowergivenaspeedinterval

changing environments.Therefore, we added adaptive variables into the fitness function:

$$F = \alpha \cdot v + \beta \cdot (P_{real} - P_{calc}) \tag{3.22}$$

Subjectto:

$$\alpha + \beta = 1 \tag{3.23}$$

In the Equation 3.22, $v$is the normalized speed, and $P_{calc}$is the normalized power.P_{real} is real-time power when calling the Genetic Algorithm.α and $\beta$are normalized weights of speed and power,respectively.More specifically,$v$can be calculated by averaging the lowerandupperboundofthespeedintheselectedchromosome:

$$v = 0.5 \cdot (v_{min} + v_{max}) \tag{3.24}$$

Likewise,forP_{calc}:

$$P_{calc} = 0.5 \cdot (P_{min} + P_{max}) \tag{3.25}$$

Moreover,theβ inEquation3.22isanadaptivepara changes withthecurrentPIDerrorwhencallingGeneticAlgorithm:

$$\beta = \beta \cdot \left(1 + \frac{Err_{PID}}{C}\right) \tag{3.26}$$

Where$C$isaconstanttocontrolthechangingmagnitudeofβ,whichcanbetunedby theuser.

3.4.2.4Crossover,Mutation,SelectionandElitism

Intheimplementation,theintermediatecrossoverisused.Itgeneratesvaluesforoff-springaroundandbetweenthevariablevaluesoftheparents.Asformutation,anor-mallydistributedrandomnumberisaddedtothegeneduringthemutationoperation, usingthefollowingformula:

$$Var_o = Var_i + \sigma \cdot N(0, 1) \tag{3.27}$$

Where σ is the standard deviation of the normal distribution; $N(0, 1)$ is the standard nor-maldistribution,withmean=0andvariance=1;σisaselectedproportionalparameter, whichisdefinedas0.2inthecurrentsettings.Forselection,thetournamentselectionis used.Severalcandidatescompetewitheachotherandthebesthavetherighttobreed. Inourcase,fourcandidatescompetewitheachotherduringtheselectionprocess.This methodguaranteestheworstindividualnevergetsthechancetoreproduce.Moreover, itisbothapplicableforminimumfitnessormaximumfitnessapproaches,whichisan advantage for testing in case we need to switch between minimum and maximum fitness. Lastbutnotleast,anelitismmechanismisimplementedinthealgorithmtoguarantee wedonotlosethebestindividual.Theelitismsimplycopiesthefittestindividualto externalstoragespaceandlateroncopiesitbacktothenextgeneration.

3.4.2.5PIDErrorCalculation

Intheimplementedalgorithm,thePIDalgorithmservestwopurposes.Firstofall,it calculatesthePIDerrorbetweenthereal-timepower(P_{real})andthecalculatedpower (P_{calc}).Iftheerrorexceedsthethreshold,thesystemwilldecidetocallanotherround ofgeneticalgorithm.ThePIDerrorthresholdiscalculatedas:

$$Err_{th-pid} = C \cdot (P_{max} - P_{min}) \tag{3.28}$$

WhereCistheusertunedparameter.P_{max}andP_{min}arethemaximumandminimum powercalculatedbythegeneticalgorithm.Whilethegeneticalgorithmisnotrunyet, theycanbecalculatedas:

$$P_{max} = f(v_{req} + 0.5 \cdot v_{int}, T) \tag{3.29}$$

$$P_{min} = f(v_{req} - 0.5 \cdot v_{int}, T) \tag{3.30}$$

Where$f$isthepolynomialregressionmodelweacquiredfromthecollectedrealmotor datathatcalculatespowergivenspeedandtorque.v_{req}isthecurrentrequestedspeed and$T$isthecurrentnormalloadtorque.Inmostthecasesinthisthesiswork,T= 0.

Secondly,iftheGeneticAlgorithmiscalled,thePIDerrorwillbepassedtotheGe-neticAlgorithmfor*beta*calculationinthefitnessfunction,asindicatedinEquation 3.26.PseudocodeforPIDerrorcalculationisprovidedinListing3.5.

```
Input:PIDparametersKp,Ki,Kd;real-timepowerP_real;calculated      ,→powerP_calc
Output:PIDerrordelta
Parameters:staticfloaterr,der_err,int_err

defPIDErrCalculation():
    #previouserror
    der_err=-err
    #currenterror
    err=P_real-P_calc
    #derivativeerror
    der_err+=err
    #integralerror
    int_err+=err

    #piderror
    delta=-Kp*err-Kd*der_err-Ki*int_err returndelta
```

Listing3.5:PseudocodeforPIDerrorcalculation

3.4.2.6GeneralProcessoftheAdaptiveControlAlgorithm

Ingeneral,theadaptivecontrolalgorithmworksasfollows:firstofall,theadaptive controlisenabledbytheservice*continuousadaptivecontrol*,asindicatedinListing 3.6.Then, the algorithm will run periodically inside a ROS loop.After that,the system willcheckthecurrentPIDerrorbetweenthereal-timepower(P_{real})andthecalculated power(P_{calc}).Iftheerrorislargerthanthepre-definedPIDerrorthreshold,theGAis called and the PID error will be passed to the Genetic Algorithm for βcalculation (as in Equation 3.26) in the fitness function.Inside the GA, the individuals go through the fol-lowingprocedures:populationgeneration,parentselection,crossover,mutation,fitness evaluationandelitism.Afteriterations,finallyanoptimalindividualisselectedandthe speed will be applied to the conveyor belt.The whole process is described in Figure 3.15.

```
rosservicecall/adaptive_control_app/continuous_adaptive_control
"{isEnabled:true,motorID:1,param1:1.0,param2:1.2,
intSpeed:8.0,Kp:2.0,Ki:5.0,Kd:1.0}"
```

Listing3.6:TheROSserviceforcontinuousadaptivecontrol

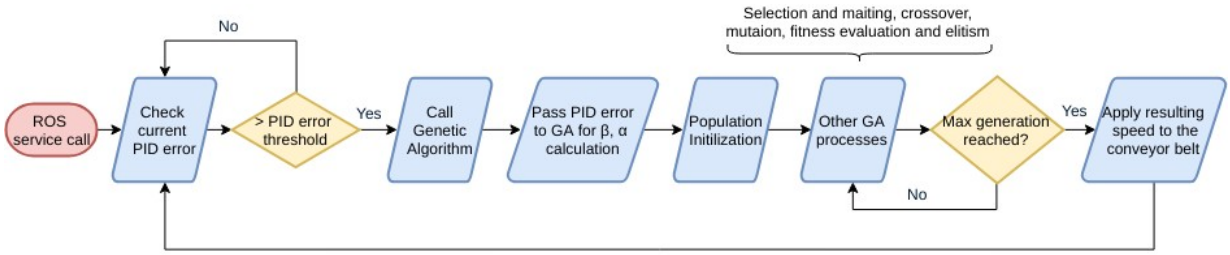


Figure3.15:Thegeneralprocessoftheadaptivecontrolalgorithm

3.5CloudAPIforIoTBasedKnowledgeManagement

Aftertheimplementationofpredictivemaintenanceandadaptivecontrolmodules,in order to make these modules work with the knowledge management service in the CHARIOT[8]project,RESTfulAPIsneedtobeimplementedforthecommunications.These APIs should provide RESTful based web access to functionality regarding database management,datastorageandretrieval,aswellasmachinelearningbaseddataanalysison the cloud.Since we are using MySQL database on the cloud, the APIs should inherently supportdataaccessviaMySQL.

3.5.1CHARIOTCloudAPI

Inthissubsection,anAPIthatprovideswebaccessestodatabaseoperations,datastorageandretrieval,aswellasmachinelearningbaseddataanalysisisimplemented.

web.pyisasimpleyetpowerfulwebframeworkforPython.Itwasfirstpublishedby AaronSwartzatreddit.com,andthewebsiteuseditintheprocessofgrowingintoone of the 1000 top websites.Basically, web.py can export the implemented Python function as a web application through several simple code lines,which is very handy and suitable fortheimplementationofourCHARIOTCloudAPI.Themostdistinguishablefeature ofthegrammarofweb.pyisthatthefunctionalityisrequiredtobeencapsulatedina Pythonclass.

RegardingthecandidatesofthedriverforMySQLdatabaseaccess,openstack.com [5]hasadetailedcomparisonbetweenthestate-of-the-artdriversforpythoninclud-ingMySQL-Python,mysqlclient,OurSQL,MySQL-Connector-PythonandPyMySQL, whichisshowninTable3.3.Thecomparisonstronglysuggeststhat PyMySQListhe bestoption.

Usingweb.pyandPyMySQL,aweb-applicationbasedAPIthathasaccesstoMySQL databaseisimplemented.FourkindsofMySQLdatabaseaccessfunctionalityissupported:databaseandtableoperations,datastorageandretrieval,otherSQLrelated operationsandmachinelearningbaseddataanalysis.ThefunctionalityoftheAPIis

Driver	PyPihosted	Eventlet friendly	Python3 compatibility	Maturity and/or stability
MySQL-Python	Yes	Partial	No	Yes
mysqlclient	Yes	Partial	Yes	Yes
OurSQL	Yes	No	Yes,butnot Pypihosted	No
MySQL-Connector-Python	No	Yes	Yes	Yes,though stillfairlynew
PyMySQL	Yes	Yes	Yes	Yes,though lackspolish

Table3.3:ComparisonsofdifferentMySQLdrivers[5]

depictedinFigure3.16.Themachinelearningrelatedcomponentswillbefurtherdis-cussedinthenextsubsection.

Allthefunctionalityrelatedtocreation,deletionandediting(i.e.writefunction)are implementedusingthePOSTmethod;whileotherfunctionalityrelatedtogettingin-formationordatafromthedatabase(i.e.readfunction)areimplementedusingGET method.ThisisbecausePOSTissafersinceitdoesnotattachinformationdirectlyto theURLs;additionally,POSThasnolimitforthesizeoftransmitteddata,thereforeit is very suitable for the function that is related to ”write”.Since the API converts all the databaseaccessfunctionalityintoRESTfulwebapplications,whentheAPIisrunning, onecaneitheruseaHTTPlibrarylikeRequestsorcommandlinetoollikecURL(GET orPOSTmethods)topassthedata,ordirectlyuseURLsthatmeetourpre-defined formats(GETmethodsonly).InListing3.7,twocodesnippetsforGETandPOST methodsareshown,respectively.

```
#GetTableData,aGETmethod
payload={}
payload["table"]="TrainingList_1"
r=requests.get(’http://YOUR_IP_ADDRESS/cloud/db/getTableData/’,params ,→ =payload)
#Alternative
http://0.0.0.0:8080/cloud/db/getTableData/?table=MotorMaintenanceList_1

#StoreDataInTable,aPOSTmethod
load={}
load["table"]="TrainingList_1"
load["path"]="training_dataset.csv"
```

```
json_load=json.dumps(load)
r=requests.post('http://YOUR_IP_ADDRESS/cloud/db/storeDataInTable/', ,→data=json_load)
```

Listing3.7:TheexamplesforGETandPOSTmethodsinCHARIOTCloudAPI

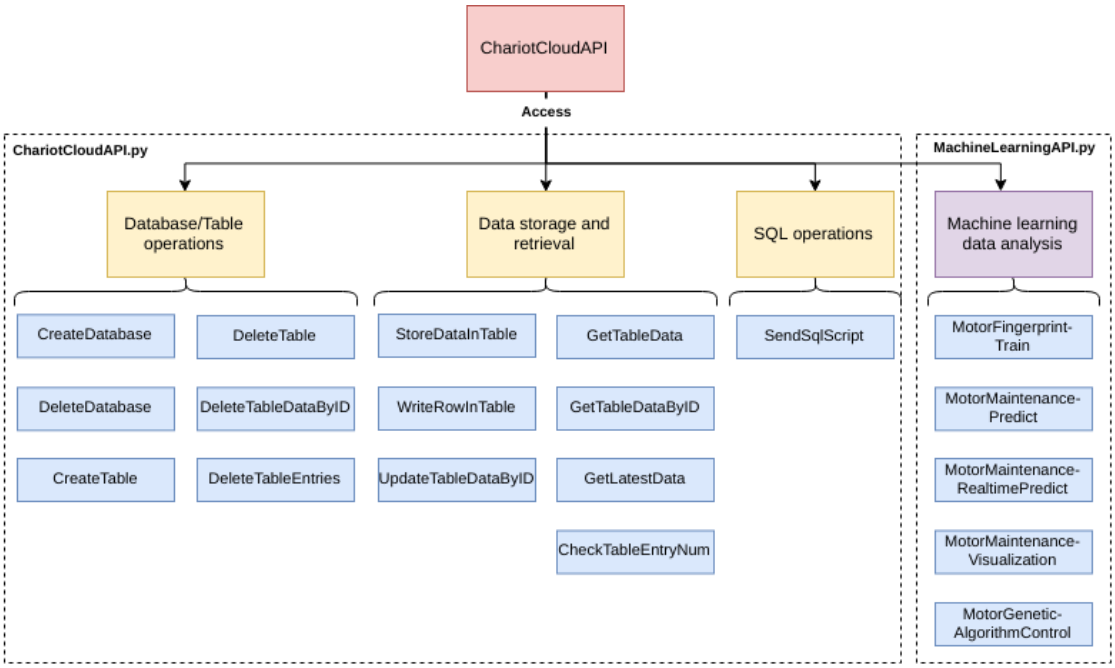


Figure3.16:ThesupportedfunctionalityinCHARIOTCloudAPI

IntheCHARIOTCloudAPIapplications,errorhandlersarealsoimplementedforthe convenienceofdebugginganddiagnosis.

3.5.2MachineLearningAPI

ThemachinelearningAPIprimarilyholdsthemachinelearningrelatedfunctionality for the predictive maintenance (class MotorFingerprintTrain, MotorMaintenancePredict andMotorMaintenanceVisualization)andadaptivecontrol(classMotorGeneticAlgo-rithmControl). ThisAPIfunctionsasasub-APIoftheCHARIOTCloudAPIand shouldbeimportedintotheCHARIOTCloudAPIasalibrary.

AstandardprocessofusingthisAPItoperformmachinelearningdataanalysisis visualizedinFigure3.17.

Specifically,inclassMotorFingerprintTrain,dataarefirstloadedfromatableofthe database;thenthemodelistrainedusingeitheroneclassSVM,*k*-NNorkernelPCA,

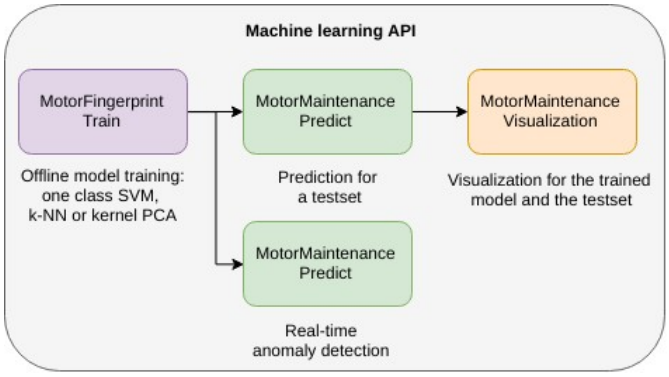


Figure3.17:ThedataflowinMachinelearningAPI

andthetrainedmodelandrelatedvariableswillbeconvertedandsavedbyPickleas SQLtablestothedatabase.InclassMotorMaintenancePredict,thesavedmodeland variableswillbereadfromthedatabaseandperformtheprediction.Thankstothena-tureofsemi-supervisedlearning,thetrainingandpredictionproceduresareseparated. Thereforethepredictioncanbefinishedinaveryshorttimeregardlesshowlongitwill taketotrainthemodel,thusitisverysuitableforreal-timeprediction.Lastbutnot least, in class MotorMaintenanceVisualization, the training and prediction results will be visualized.However,sincecalculatingthedecisionboundaryofthenormaldatapoints isusuallytime-consumingprovidedthatthetrainingdatasetsizeislarge,visualization duringrun-timeishighlynotrecommended.

4ExperimentandEvaluation

Thischapterpresentshowtheexperimentsareconductedinthisthesiswork.Thema-jorpurposeofconductingtheseexperimentsistoevaluatehowtheproposedpredictive maintenanceandadaptivecontrolalgorithmsperformunderdifferentparametersand howmuchtheyimprovecomparingtotherule-basedmethods.

Thechapterisseparatedintotwosubsections.Thefirstsubsectionintroducestheex-perimentsettingsandthesystemarchitectureofthesystem.Thesecondoneincludes descriptionsofhowtheexperimentsareconducted,theexperimentresultsandthecor-respondinganalysis.

4.1ExperimentsetupandSystemArchitecture

4.1.1SystemArchitecture

The proposed mechanisms are parts of the CHARIOT [8] Project.In order to communi-cate with the CHARIOT Middleware,a special system architecture was implemented as inFigure4.1.Generally,mostofthesystemisrunningontheROS.Thephysicallayer directlydealswiththeservomotorandtherobotarm.Themiddlewarelayerserves asabridgebetweenthephysicalandapplicationlayer.Itintegratesandencapsulates theROSservicesandtopicsfromthephysicallayerandforwardsthemtotheapplica-tionlayer.Additionally,adatagatewaywasimplemented(whichwillnotbediscussed inthisthesiswork)toforwardtheROSservicesandtopicstotheKnowledgeManage-ment Agent in the CHARIOT Middleware so that agents in the ROS environment (lower layers, including physical layer) can exchange information (e.g.sending sensory informa-tion,receivingcontrolsignals)withCHARIOTagents.Regardingtheapplicationlayer, itdealswithmorespecifictasksdefinedbytheusers.Moreover,theChariotCloudAPI providesRESTfulaccesstotheMySQLdatabaseandthemachinelearningalgorithms.

4.1.2Experimentsetup

4.1.2.1Hardware

Inthisthesiswork,atestbenchincludingaconveyorbelt,arobotarm,aninfrared sensor,acolorsensor,acurrentsensor,twoRaspberryPi3boards,andvariousother components,wasusedfordatacollectingandexperimentsasshowninFigure4.2.

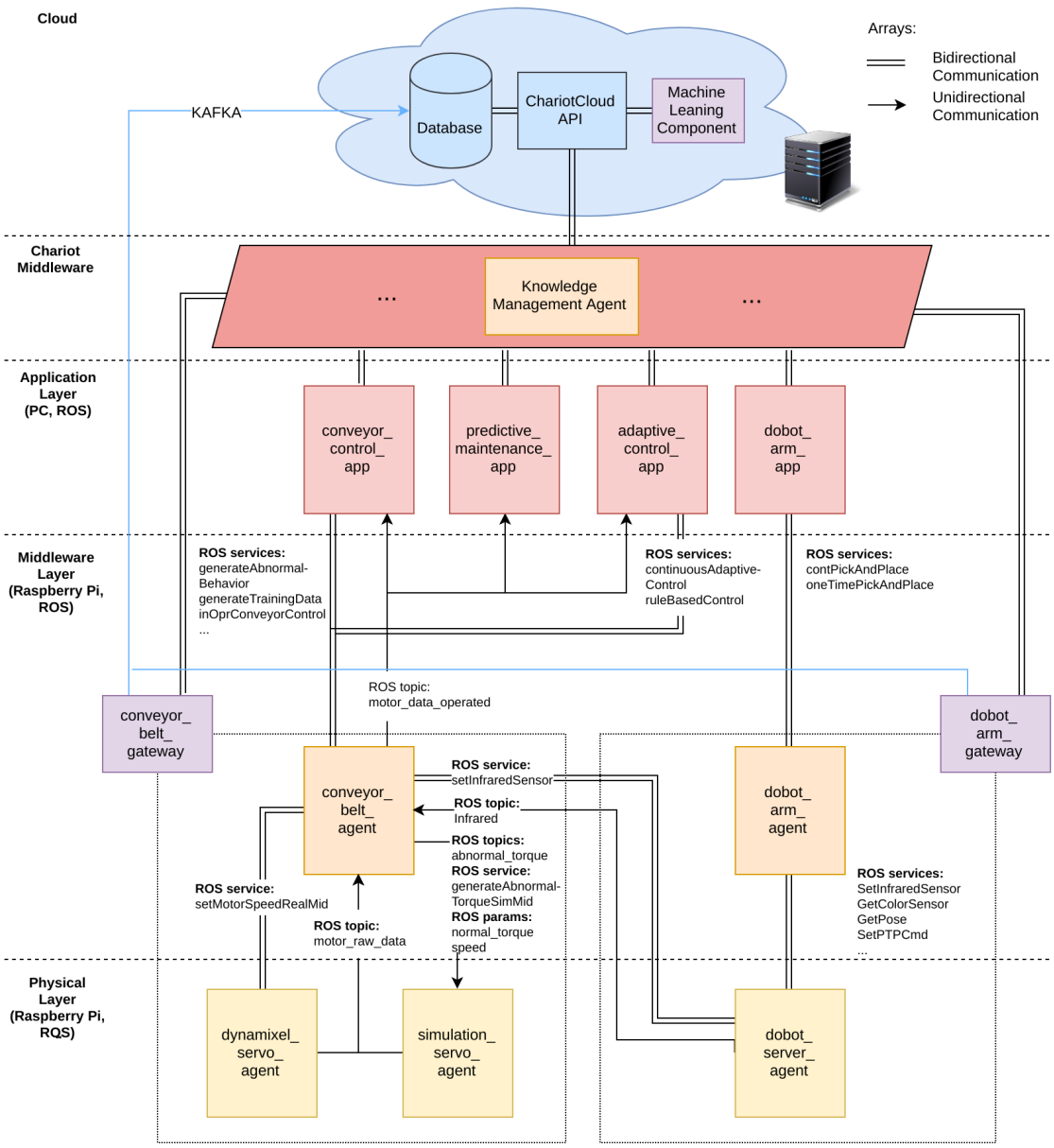


Figure4.1:Thearchitectureofthesystem

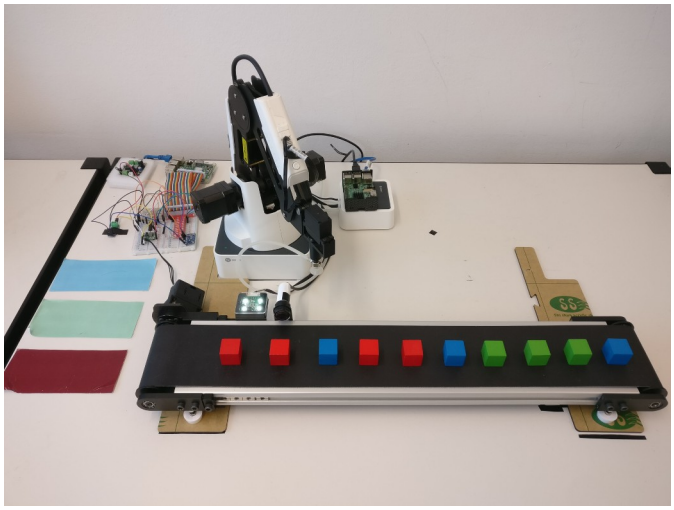


Figure4.2:Thetestbenchusedintheexperiments

ArobotarmfromDobotcalledMagician[71]wasused.Additionally,tosimulatethe realenvironmentinthefactories,aconveyorbeltkitincludingaconveyorbelt,anin-fraredsensorandacolorsensorwasalsoused.Thecontentsoftherobotarmandthe conveyorbeltkitareshowninFigure4.3.

However, the original stepper installed on the conveyor belt had limited control accuracy andcannotprovideanyreadingsaboutthemotor.Therefore,itwasreplacedbythe servomotorXL430-W250fromDynamixel,whichsupports360◦rotationina0.088◦resolutionandvariousreal-timereadings,includingvoltage,PWM,speed,temperature, etc.The servo motor is shown in Figure 4.4.As the connection components between the originalconveyorbeltandthesteppermotorwerehighlycustomized,wecannotsimply replacesteppermotorwiththeservomotor.Instead,aspecialmechanicalinterfacewas designedand3Dprinted.TheMechanicalInterfacefortheservomotorisshownin Figure4.5.

Inordertomaketheconveyorbelt-robotarmsystemwork,threeadditionalproblems neededtobesolved.Onewasthecommunicationproblembetweenthemotorandthe RaspberryPi.TheDynamixelServoMotorXL430-W250usesaTTLhalf-duplexasyn-chronousserialcommunicationprotocol,whileRaspberryPi3onlyhasafull-duplex UARTchannel.Thereforeanintegratedcircuitchipcalled74LS241wasusedtofin-ishtheconversion.Secondly,tocalculatethepower,sincecurrentreadingswerenot providedbytheservomotor,anadditionalcurrentsensorwasrequired.Inourcase,a bidirectional current sensor ACS712 was used.But ACS712 only provides analog voltage readingsthatareproportional(140mVperA)tocurrentandthereforeananalog-to-

Figure4.3:ContentsoftheDobotrobotarmandtheconveyorbeltkit[3]

Figure4.4:ServomotorDynamixelXL430-W250fromRobotis[4]

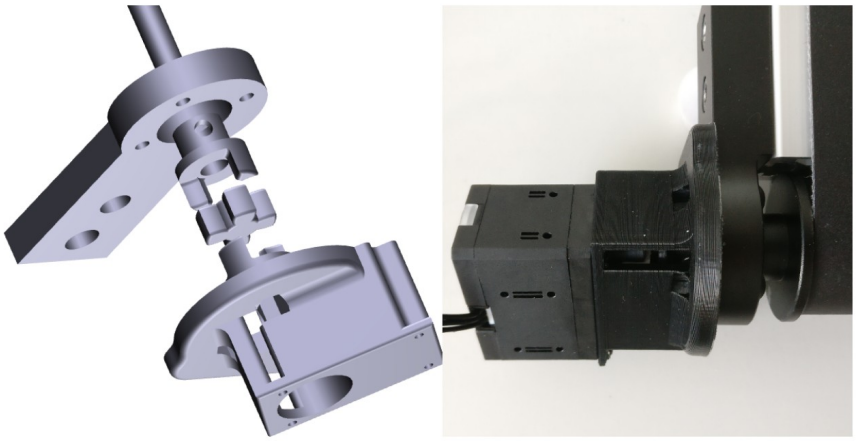


Figure4.5:Mechanicalinterfacebetweentheservomotorandtheconveyorbelt

digital converterchip ADS115 was used to generatedigitalsignalsfor the RaspberryPi. Lastbutnotleast,regardingthepowersupplyissues,inordertoprovide11.1Vtothe servomotorand5Vtoothercomponents,a12VpowersupplyandtwoDC-DCvoltage converterswereused.Theelectricalconnectionofthesystemisindicatedasin Figure 4.6.

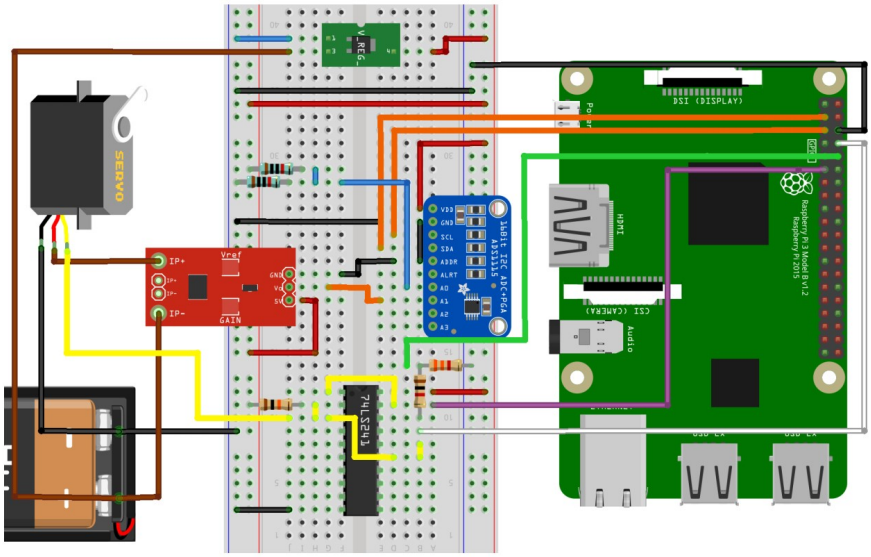


Figure4.6:Theelectricalconnectionsofthesystem

4.1.2.2Software

Inthisthesiswork,mostoftheimplementationsandexperimentswereconductedon aPCwithIntel(R)Core(TM)i7-4710MQCPU@2.50GHz,16GBRAMandUbuntu 16.04.5LTS.TheVisualStudioCodeversion1.26.0andtheRoboWareStudiover-sion1.1.0wereusedasdevelopmentIDEs. Python,C++wereusedduringthede-velopmentandexperiments.LibrariessuchasROS[72],Scikit-learn[21],NumPy[73], matplotlib[74], JSON[75], web.py[76], Requests[77], PyMySQL[78], Gaft[79], Dobot API[80], etc.werealsoused.

Additionally,whengeneratingthesimulationmotormodel,Matlab2017bSimulinkwas usedonWindows10.

Asforthecloudpart,theMySQLdatabaseandtheChariotCloudAPIwererunon aUbuntu16.04.5LTSvirtualmachineonaserverwithfourIntel(R)Xeon(R)CPU X7550@2.00GHz,32GBRAM.

4.2Experiments

4.2.1PredictiveMaintenance

In the previous chapter, we refer to three anomaly detection algorithms that are suitable forpredictivemaintenance.Eachofthemhasseveralkeyparametersthatwillhighly affecttheperformanceofthealgorithm.Inthissubsection,wewillfirstselectthebest keyparametercombinationforeachcandidate,thencomparethreecandidatestoselect thebestone.Later,thepredictivemaintenancealgorithmisimplementedusingthe best candidate with the best parameters.Finally,our algorithm is compared with a rule basedalgorithmtoshowtheeffectivenessofpredictivemaintenanceinsuchoperations. Notethatinalltheexperimentsconductedforpredictivemaintenance,asimulation motorwithrealisticnoisesisused.Theservomotorinourexperimentsetupsuffers fromtheimbalancedfrictionontheconveyorbeltandinaccuracyofthecurrentsensor. Itmightobtainunexpectedreadingssometimes,thereforeitisnotconsideredasthe optimalchoice.

4.2.1.1ParameterSelectionsforDifferentAnomalyDetectionAlgorithms

Generationofartificialoutliersandpseudotargetdata

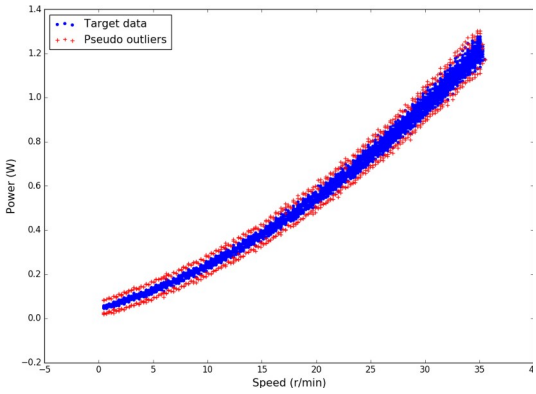
Thegeneralwayofdoingparameterestimationisusinggridsearchwithcrossvali-dation.Thecrossvalidationhelpsustofigureouthowthealgorithmperformswhen therearenewdata.Itcanalsoeffectivelypreventthetrainedmodelfromover-fitting. However,theessentialproblemhereisthatthenatureofanomalydetectionproblems istheabsenceoftheoutlierclass.AsdescribedinSection3.3.1.1,weusethemethod”negativeshifting” proposedbyWangetal.in[2]togeneratetheartificialoutliers.

Moreover,ascrossvalidationistime-consuming,Wangetal.alsoproposedthe”positiveshifting”methodtogenerate”pseudotargetdataset”formodelvalidation.The overallerrorrateoftheevaluatedmodelcanbedirectlycalculatedas:

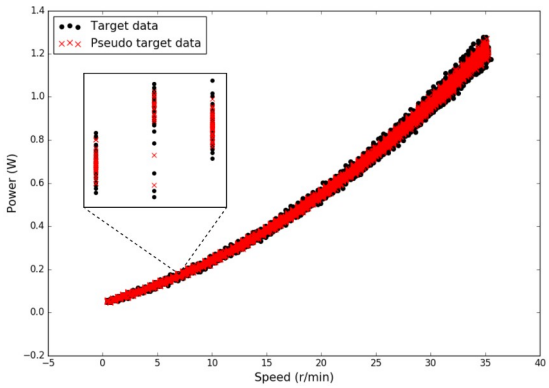
$$Err = 0.5 \cdot Err_o + 0.5 \cdot Err_t \tag{4.1}$$

WhereErr_oistheerrorrateontheartificialoutlierclassandErr_tistheerrorrateon thepseudotargetclass.

Usingthemethodabove,theartificialoutlierdatasetandthepseudotargetdataset weregeneratedasinFigure4.7.Sinceallofthethreealgorithmsareunsupervised,we canusethesetwodatasetsforallthealgorithmcandidates.



(a)Pseudooutliers



(b)Pseudotargetdata

Figure4.7:Pseudooutliersandpseudotargetdata

ParameterselectionforoneclassSVM

AsdescribedinSection3.3.1.1,inoneclassSVM,wearemainlyinterestedintwo parameters:thehyperparameterνfortheoneclassSVMalgorithmandthehyperpa-rameterγfortheRBFkernel.Theinfluenceofthesetwoparameterstothemodelis alsodiscussedinSection3.3.1.1.Usingthemethodmentionedabovecombiningwitha gridsearch,theoverallerrorratecanbecalculated.TheresultsareshowninTable4.1.

Thebestparametersare:ν= 0.08, γ= 15.Thecorrespondingtrainedmodelisdemon-stratedinFigure4.8.Fromthegraph,wecanseethatthedecisionboundarytightly surroundsthetrainingdata.Mostofthegeneratedartificialoutliersarepredictedas abnormalandmostofthepseudotargetdataarepredictedasnormal.Buttheresults alsoshowthat,underthisparametersetting,oneclassSVMiseasilyaffectedbythe trainingdata.Somejaggededgesalreadyappearonthedecisionboundaryduetothe fluctuationsofthedata.Thiswillbebeneficialifwearecertainthetrainingdataare completelynormal.Otherwise,ifthetrainingdataarecontaminated,theaccuracyof themodelwouldbeundermined.

ν \ γ	2	5	8	10	15
0.01	4.803	4.228	2.901	2.786	1.368
0.02	4.354	3.042	2.476	1.950	1.281
0.05	2.820	1.128	0.941	0.810	0.865
0.08	2.640	0.621	0.569	0.459	0.372
0.1	3.345	1.023	0.763	0.468	0.425

Table4.1:Theerrorrate(%)ofoneclassSVMfordifferentνandγcombinations

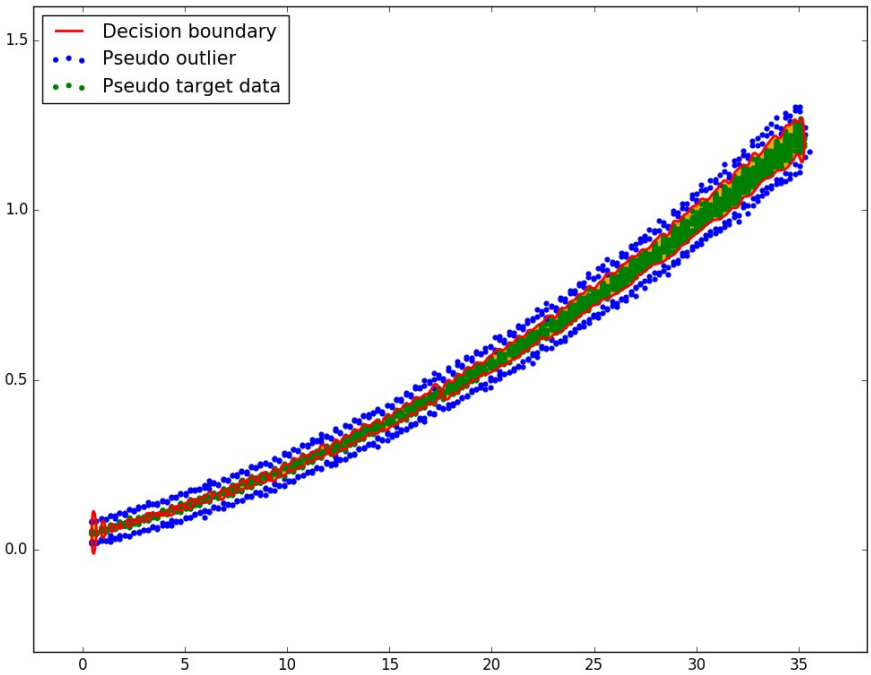


Figure4.8:TheoneclassSVManomalydetectionmodelwhenν= 0.08, γ= 15

Parameterselectionfor*k*-NN

Similarly,in*k*-NN,wemainlycareabouttwoparameters:thenumberofnearestneigh-bor *k*used by k-neighbors queries and the leaf size *leafsize* of the constructed BallTree or KDTree.Research regarding the influence of *k*and *leafsize* are discussed in Section 3.3.2.1.

However,fromtheresultsof*k*-NN,wenoticethatthedecisionboundaryistoofar awayfromthetrainingset,thusitalwayspredictsmostoftheoutlierswrong(false positive),whilemostofthetargetsright(truepositive).Inthiscase,its$Err \approx 0.5$for

allthe parametercombinationsasindicatedinTable4.2.Thereforeitshouldnotbe selected as the best algorithm.A trained model with *k*= 3 and *leafsize* = 30 is shown inFigure4.9.Resultsshowthatallthe artificialoutliersandpseudotargetdatawere predictedasnormal.

ν \ γ	5	10	20	30	50
3	49.923	49.923	49.923	49.923	49.923
5	49.923	49.923	49.923	49.923	49.923
8	49.923	49.923	49.923	49.923	49.923
10	49.923	49.923	49.923	49.923	49.923
15	49.923	49.923	49.923	49.923	49.923

Table4.2:The error rate (%) of one class SVM for different *k* and *leafsize* combinations

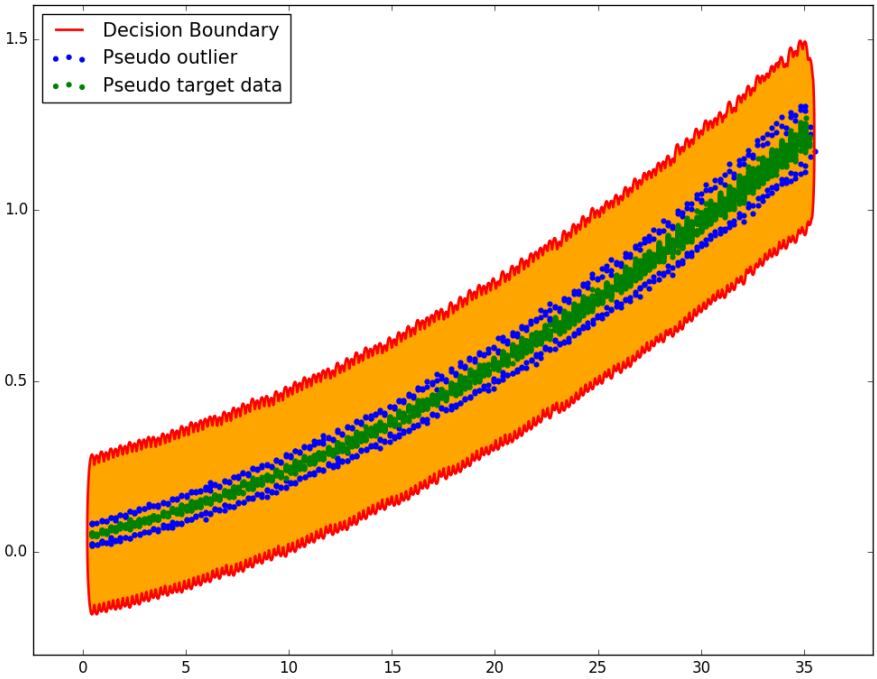


Figure4.9:The*k*-NNanomalydetectionmodelwhen*k*= 3and*leafsize* = 30

ParameterselectionforkernelPCA

InkernelPCA,wearemainlyinterestedintwoparameters:thehyperparameterα =of theridgeregressionthatlearnstheinversetransformandthehyperyparameterγofthe RBFkernel.Researchesregardingtheinfluenceofα=andγarediscussedinSection

3.3.3.1.Theparameterselectionresultsareshowninn Table4.3.

α \ γ	2	5	8	10	15
0.005	40.261	14.724	9.279	6.529	2.542
0.01	40.261	18.788	10.199	8.589	3.615
0.02	40.721	23.006	12.653	10.123	4.842
0.05	42.868	28.604	16.641	13.727	8.282
0.1	46.012	32.592	20.706	17.255	11.426

Table4.3:Theerrorrate(%)ofkernelPCAfordifferentαandγcombinations

Thebestparametersareα = 0.005, γ= 15.Thecorrespondingtrainedmodelisdemonstratedin Figure4.10.Thegridsearchalgorithmfavorsbiggerγandsmallerαtoget betterperformancearoundthedecisionboundary. Thesideeffectofthisparameter combinationisthat,somejaggededgesalreadyappearonthedecisionboundary.

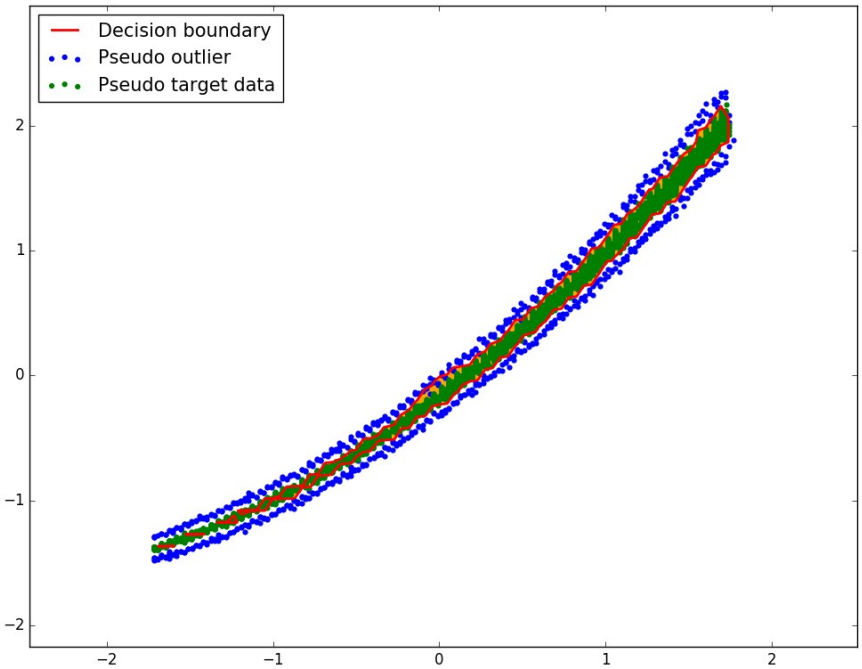


Figure4.10:ThekernelPCAanomalydetectionmodelwhenα = 0.005, γ= 15

4.2.1.2ComparisonsbetweenDifferentAnomalyDetectionAlgorithms

Inthispart,wecomparethreecandidatesusingtheirbestparameterstoselectthebest algorithm.Sincewareusingthesameartificialoutlierandpseudotargetdatasetsfor allthealgorithms,wecandirectlycomparetheirperformancethroughthecalculated errorrateandthetraining/predictingtime,theresultsareshowninTable4.4.

Items	Error Rate(%)	Training Time(s)	PredictingTimefor
oneclass SVM	0.372	0.170	0.062
k-NN	49.923	0.0034	0.538
kernel PCA	2.542	65.281	1.758

Table4.4:TheperformancecomparisonofoneclassSVM,*k*-NNandkernelPCA

Theresultsshowthat,firstofall,*k*-NN'serrorrateisremarkablyhigh,predictingall theoutliersaswithinnormalrange,thusnotusableinpractice.Furthermore,although kernelPCAalsogetsrelativelygooderrorrate,ittakesconsiderabletimetofinishthe trainingaswellastheprediction.Thisperformanceisapparentlynotsuitableforreal-timeapplications.OneclassSVMnotonlyhasthesmallesterrorrate,butalsohas muchshortertraining/predictingtimethankernelPCA.Thereforeweselectoneclass SVMasourpredictivemaintenancealgorithm.

4.2.1.3ComparisonsofthePredictiveMaintenanceAlgorithmwithaRule-based Algorithm

Inthispart,usingoneclassSVMwiththebestparametercombinations(v= 0.08, γ= 15),areal-timeprogramtodemonstratethefunctionalityofpredictivemaintenanceis implemented.Theprogramisshowingreal-timeresultsinaspeed-powergraph,where notonlythecollectedhistoricaldata,butalsothenewobservationsofthemotorare plottedonthegraphasdatapoints.Thenormaldatapointsareshowningreencolor, whiletheabnormalonesareshowninredcolor.Themovingaverageisappliedtoalle-viatethepowerreadingfluctuationscausedbynoises.Inthebackground,theprogram will send queries to the cloud API to request real-time classification results based on the already trained one class SVM model.Due to the delay effect caused by moving average calculation,afterthealgorithmdecidesseveralconsecutiveobservationsasabnormal,a predictivemaintenancereminderwillbeshownonthescreen;inthemeantime,aROS topicisalsopublishingthereal-timeclassificationresultsinthebackground.

ThefollowinggraphsinFigure4.5showthedecisionboundaryoftherule-basedal-gorithm.The rule-based mechanism has a ladder shape boundary.The motor operating speedrangeisdividedintosevenintervals:[0,5],[5,10],...,[30,35].Ineachinterval,1.2

timesthebiggesthistorypowervalueattherightendpointoftheintervalatisusedas thepowerlimit.Forexample,for[30,35]interval,thepowerlimitwillbecalculatedas $P_{[30,35]}$= 1.2·P_{max}(v= 35), where $P_{max}$is the power limit and $v$is the speed.This is to simulatethesituationsinthe realworld,whereonlypowerlimitsatsomecritical points areknown.Thereasonwhya20%slackisreservedisthat,consideringthelimited observationdata,inrulebasedsystem,wehavenoabsoluteconfidencethattheknown maximumpoweristheupperboundofthenormaloperationrange.

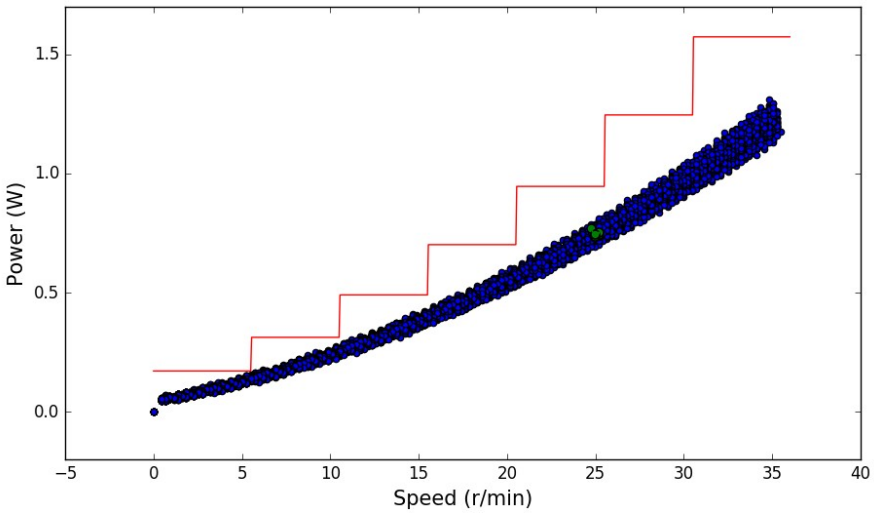


Figure4.11:Decisionboundaryforrulebasedmechanism

Intheexperiments,atorquechangingwithtime$T$=0.002 · Δ$t$isappliedtosimulate theabnormaltorque.Theaveragetimebeforegettingpredictivemaintenancereminder isusedasametrictomeasuretheperformanceofthealgorithm.Moreover,twospeed values22and25r/minareselected,becauseintuitively,speedthatisclosertothe bottomoftheladderreactsslower.Wewouldliketostudytheeffectofthat.The resultsareshowninTable4.5.

Speed (r/min)	Avg.TimeforPredicted Maintenance(s)	Avg.Timefor Rulebased(s
22	6.165	35.29
25	6.412	20.80

Table4.5:Comparison of average waiting time for predictive maintenance and rule based mechanisms

Theresultsshowthat,firstofall,therule-basedsystemhasamuchlongerwaiting

time than the predictive maintenance algorithm.Secondly, the rule-based system suffers **form a large waiting time difference between the speed = 25 r/min and speed = 22 r/min** situations.While this is not the case in our predictive maintenance algorithm.Therefore, itisnotablethatintherule-basedmechanismifthespeedisclosetotheleftendpoint oftheselectedspeedintervals,itmighthasahigherchancetogetdamagedbefore thepredictivemaintenancereminderissent. Ingeneral,ourpredictivemaintenance algorithm is better in terms of reaction speed and the ability to protect the components.

4.2.2AdaptiveControl

Inthissubsection,weevaluatehowtheintervalsizeandtheweightsofpowerandspeed inthefitnessfunctioninfluencetheperformanceoftheproposedadaptivecontrolalgo-rithm.

Firstofall,consideringtheabnormalconditionsthatmayhappeninarealindustrial conveyorbelt,forexample,extraperiodicfrictionsthatoccurwhenthecomponentsare wornout,theabnormaltorquementionedinSection3.2.1.2wasusedtosimulatethese phenomena.Alsoconsideringthefactthattherealmotoriseasilyaffectedbyextra factorssuchasunbalancedfrictionsontheconveyorbeltandinaccuracyofthecurrent sensor,inthisexperiment,onlythesimulationmotorisused.Moreover,theidealsimu-lationmodelwithoutanynoiseisusedtobetterassesstheinfluenceofparameters(i.e. intervalsizeandtheweightsofpowerandspeed).Additionally,aspeedrangeof6to 32r/minwasusedintheexperimentsasthiswasthemostfrequently-usedspeedrange inwhichourrealmotorsetupcouldsafelyoperate.Inthisway,wecreateanexact simulation of our real motor setup, with the advantages of removing any noise for better fine-tuningoftheparameters.Intheend,theadaptivecontrolalgorithmwiththebest parametercombinationwillbecomparedwiththerule-basedmechanismusingrealistic noisysimulationmotortodemonstratetheeffectivenessofouralgorithm.

4.2.2.1Theeffectofdifferentintervalsizesontheperformance

This experiment is intended to evaluate the performance of different size of intervals.For that purpose, we propose a customized efficiency calculation having both power (energy) andspeed(distance)astheevaluationmetrics.Itisdefinedasfollows:

$$\eta = \left(1 - \frac{E_\Delta}{E}\right) \cdot \alpha + \frac{S_{AC}}{S_{req}} \cdot \beta \tag{4.2}$$

Whereηistheefficiency,αandβaretheweightsthatarethesameastheweightsin the fitness function Equation 3.22.$E_\Delta$is the absolute deviance between the energy con-sumptionunderconstantrequestedspeedandtheenergyconsumptionunderadaptive speed control.$S_{AC}$is the travelled distance of the conveyor belt under adaptive control. Specifically,sincethereal-timepowerconsumptionischangingallthetime,$E_\Delta$canbe

calculatedbytheintegralofabsolutepowerdifferenceovertime:

$$E_{\Delta} = \int_{t_0}^{t_0+\Delta t} |P_{AC} - P_{req}|\, dt \quad (4$$

SimilartoE_{Δ},S_{AC}canbecalculatedbytheintegralofreal-timespeedovertime:

$$S_{AC} = \int_{t_0}^{t_0+\Delta t} v_{AC}\, dt$$

E_{req}andS_{req}aretheenergyconsumptionandthetravelleddistanceunderrequested speed,respectively.Theycanbecalculatedas:

$$E_{req} = P_{req} \cdot \Delta t \quad (4.5)$$

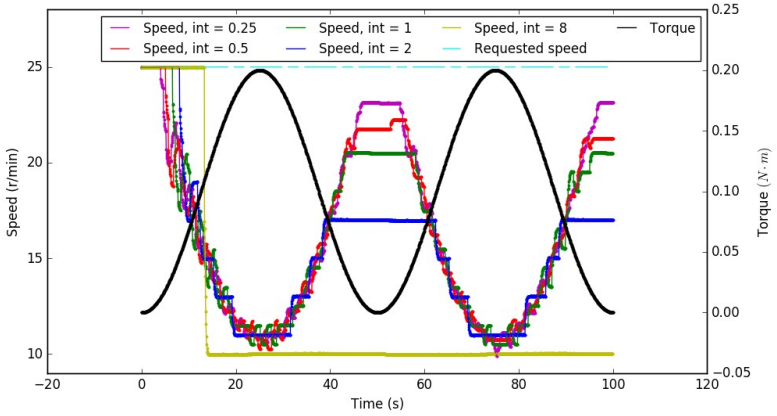
$$S_{req} = v_{req} \cdot \Delta t \quad (4.6)$$

Inourcase,agoodadaptivecontrolalgorithmshouldhavegoodflexibilityandgood smoothness.Flexibility,asitsnamesuggests,ishavingareliableresponsenotonlyto theloadtorqueincreasebutalsototheloadtorquedecrease.Inotherwords,when thedisturbance,abnormaltorqueinourcase,iswithdrawn,thecloserthespeedtothe originallyrequestedspeedis,thebetter.Smoothness,ontheotherhand,representsthe smoothnessofthespeedcontrolwhentheloadtorqueincreasesordecreases;inother words,thelesssuddenchanges,thebetter.Thesetwometricscannotbemeasured, however,canbevisualizedovertheperformancegraph.

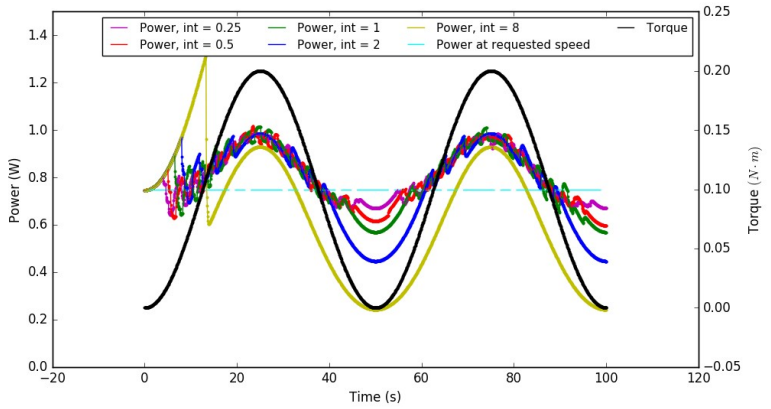
Wetestedthealgorithmforspeedintervalsof0.05,0.25,0.5,1,2and8r/min,re-spectively.Therequestedspeedwas25r/min.Tobetterevaluatethecharacteristicof thesystem,agraduallychangingsinusoidalabnormalloadtorquewithperiod$T$= 50s, runningtime$t$=100s,amplitude$A$=0.1wasapplied.Ourintentionistosimulate periodicfrictionthatcanhappenontheangularmotionofthemotor,e.g.onacertain partofthegear,oronthesurfaceoftherotatingconveyorbelt.Theexperimentresults (speed, power, and efficiency) of different sizes of intervals are shown in Figure 4.12.The correspondingaverageefficiencycalculatedbyequation4.2ofdifferentsizeofintervals areshowninTable4.6.

Interval size	0.05	0.25	0.5	1	2	8
Efficiency	0.7775	0.7747	0.7665	0.7547	0.7159	0.5930

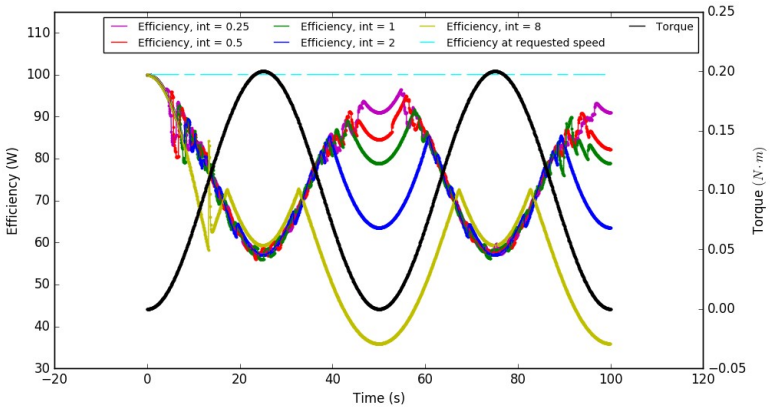
Table4.6:Theefficiencycomparisonsfortheadaptivecontrolalgorithmfordifferent intervals



(a)Thecomparisonofspeedamongvariousintervals



(b)Thecomparisonofpoweramongvariousintervals



(c)Thecomparisonofefficiencyamongvariousintervals

Figure4.12:Performancecomparisonsamongvariousintervals

	0.05vs0.25	0.25vs0.5	0.5vs1	1vs2	2vs8
df(betweengroup	1	1	1	1	1
df(withingroups)	1788	3582	3613	3688	3815
F	0.0859	3.9143	7.6002	131.8707	791.3422
P	0.7695	0.0480	0.0059	5.1397E-30	2.1979E-158

Table4.7:TheANOVAanalysisoftheefficiencyforneighboringintervals

From the experiment results presented in Table 4.6 we can conclude that the smaller the intervalsizeis,thehighertheefficiencyis.However,theinterval0.05,0.25,0.5and1, especiallyinterval0.05and0.25,donothavesignificantdifferencesintermsofaverage efficiency.ThereforeweconductedANOVAforeachtwoneighboringintervalsandset theANOVAparameterα = 0.05.TheANOVAresultsareshowninTable4.7.There-sultssuggest,thereisnosignificanteffectofspeedintervalonefficiencyatp<0.05level for0.05vs0.25:[F(1,1788)=0.0859,p=0.7695].Onthecontrary,thereisasignificant effectofspeedintervalonefficiencyatp<0.05levelforeveryotherneighboringgroup. Thesimilarityonefficiencyongroup0.05and0.25canbeexplainedbythegenetic algorithmcomputationbottleneck.Wheninterval=0.05,thePIDerrorthreshold(as inEquation3.28)isalsosmaller,whichleadstomorefrequentgeneticalgorithmcalls. However,inpractice,thecalculationofgeneticalgorithmtakestimeanditisnotfast enoughforinterval=0.05.

Furthermore,fromFigure4.12awecanseethat,firstofall,groupswithbiggerinterval sizereactslowerthangroupswithsmallerintervalsize.Thisisbecauseofthedefinition ofPIDerrorthreshold:largerintervalsresultinalargerthreshold,thereforelongerre-actiontime.Secondly,thespeeddecreasingrateisalmostthesamefordifferentgroups. Thisisbecausewhenthespeedisdecreasing,betweeneachgeneticalgorithmcalls,the calculatedPIDerrorisusuallymuchlargerthanthethreshold.ThereforethenextGA isalmostimmediatelycalledevenforlargerintervals.Thirdly,thespeedofgroupswith smallerintervalsexhibithigherflexibility,i.e.itusuallygoeshigherwhenthetorque reachesamaximum.Thereasonissimilartothefirstconclusion.Smallerintervals leadtoasmallerPIDerrorthreshold,henceit’shardertokeepthecurrentspeedeven when: (1)theloadtorquechangesmildly;(2)theGAgetscalledtheleastfrequently (whichhappensexactlyatthepeakofthesinusoidalcurve).Wheninterval=0.25,the algorithm exhibits the best flexibility, its speed returns to around 23.375 r/min when the torqueiscompletelywithdrawn.Asacomparison,wheninterval=8,afterdecreasing the speed, it is never able to return to a higher speed.Last but not least, the smoothness increases when the speed interval size becomes smaller.This is because smaller intervals makethegeneticalgorithmfunctioncallmoreoften,hencethealgorithmismorelikely toselectanearbyspeed,ratherthanafarawayone.

Asforpower,fromFigure4.12bwecanseethatthegroupswithsmallerintervals decreasetheirspeedfasterwhentheabnormaltorqueisapplied,andreturncloserto

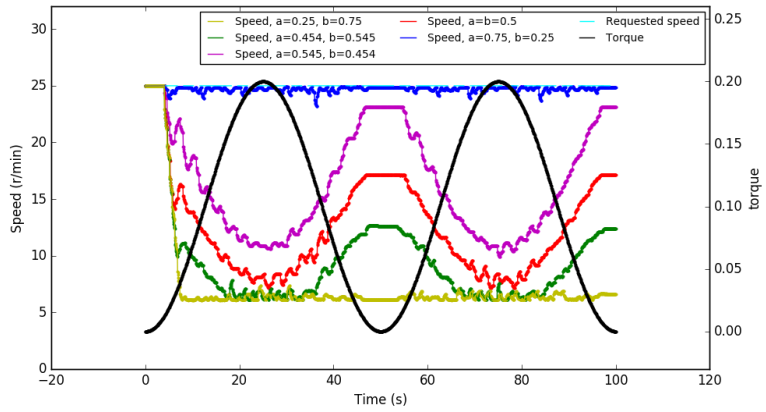
the requested speed after the torque is withdrawn; therefore, they have the closest power consumptionbehaviourastheoneatrequestedspeed.Moreover,inFigure4.12c,the efficiencyresultscombinefeaturesfrombothspeedandpower.Groupswithsmaller intervalsexhibithighervariancebutbecomeclosertotheefficiencyatrequestedspeed. This is in line with the fact that they achieve higher average efficiency.In general,when interval=0.25,wegetthebestperformance.

4.2.2.2TheeffectofFitnessFunctionWeightsonthePerformance

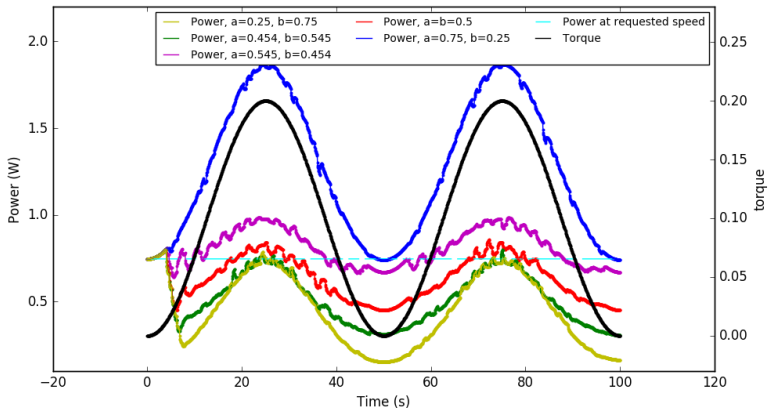
Inthispart,weevaluatetheeffectofdifferentfitnessfunctionweightsα(forspeed) andβ(forpower)ontheperformanceofthesystem.Basedonthefitnessfunctionin Equation3.22,togetafitnessaslargeaspossible,thealgorithmwilltendtoselectan individualwithhigherspeedaswellaslargerpowerdifference,i.e.chromosomeswith higherspeedandsmallerhistoricalpowerreadings.Therefore,differentcombinations ofαandβcanlargelyaffecttheperformanceofthesystem.Notethatasdefinedin Equation3.23,αandβarenormalized,thereforeactuallyonlyoneparameterneedsto betuned.Hereweusethebestintervalwegetfromthelastpart,whichis0.25.The comparisonresultsofdifferentαandβareshowninFigure4.13. Thespeedbehaviourresultsin Figure4.13ashowthat:firstofall,mostofthespeed curves roughly have mirroring shapes as the torque curve,which indicates the algorithm cancontrolthemotorspeedtoreactagainstthetorquechanges.Secondly,thecombi-nationofαandβcontrolsthehighestandlowestspeedthealgorithmcanreachwithin onerun.Thesmaller$\alpha$is,thelowerspeedthealgorithmcanreachwhentheapplied loadtorqueisatitsmaximum;whilesmallerαwillalsounderminetheabilityofthe algorithmtogobacktotherequestedspeedaftertheappliedloadtorqueiswithdrawn. Thiscanbeverifiedbythecurvewithα=0.25andα=0.75,theyreachthelower andupperlimitationsofthespeedrange,respectively.Finally,wecanconcludethat,in ordertogetthebestcontrolrangeandavoidsaturationatthespeedrangeboundaries,α = 0.545wouldbethebestchoice.

As for power consumption, results in Figure 4.13b indicate that, first of all, although the power curvesstill havea similarshape asthetorque curve,ouralgorithmcanneutralize mostofpowerincreasecausedbytheappliedabnormaltorque.Secondly,thesmaller$\alpha$is,thesmallerth epowerwillbeduringtheoperations.However,sincewedon’twant tolosemuchoperationefficiencywhileperformingtheadaptivecontrol,wearenotpur-suinglesspower,butpursuingtheproximitytothe”requestedpower”(i.e.thepower atrequestedspeedwithnoabnormaltorqueapplied).Eitherαbeingtoohighortoo lowwillleadtolargedeviationfromthe”requestedpower”,thereforeshouldnotbe selected.Finally,itissafetoconcludethat,α = 0.545, β= 0.454isstillthebestchoice consideringthatitisnotonlyclosesttothe”requestedpower”,butalsohasthemost stableperformance.

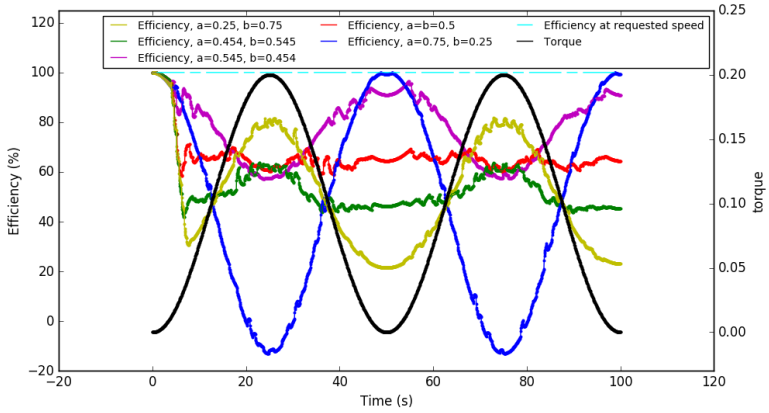
Regardingefficiency,asitisinherentlydefinedtocombinethefeaturesofpowerand speedperformance,theresultsinFigure4.13cshowtheeffectsofαandβonspeed



(a)Thecomparisonofspeedamongvariousαandβcombinations



(b)Thecomparisonofpoweramongvariousαandβcombinations



(c)Thecomparisonofefficiencyamongvariousαandβcombina- tions

Figure4.13:Performancecomparisonsamongvariousαandβcombinations

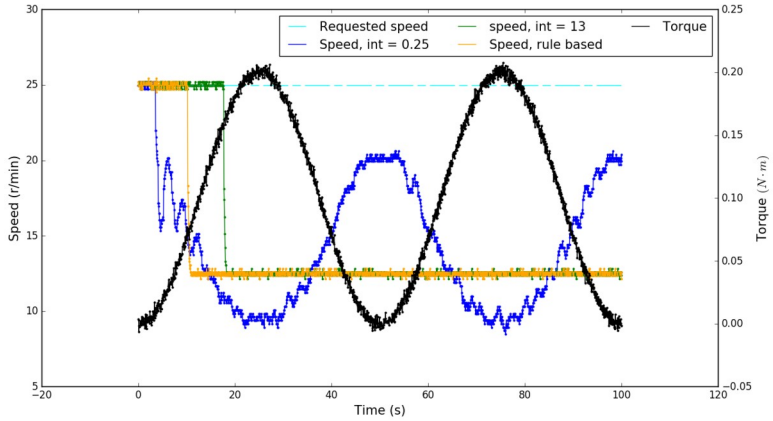
andpowercomprehensively.Fromthegraph,wecanseethatα = 0.545andβ= 0.454 givesthesmallestvariance,aswellasthehighestefficiency,thereforeitistheoptimal selectionforparameters.

4.2.2.3ComparisonswithRule-basedMechanism

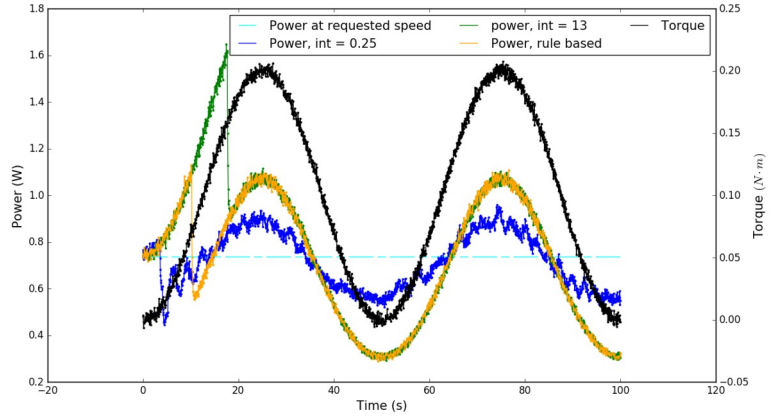
Inthispart,theresultsunderthebestsettings(*int*=0.25, α=0.545, β=0.454,i.e. AC(int=0.25))arecomparedwiththerule-basedmechanismonarealistic(noisy)sim-ulationmotortoshowtheeffectivenessofouralgorithm.Therule-basedmechanism, asitsnamesuggests,slowsdownthemotorspeedbasedonacertainrule;whichinour experimentsettings,isdefinedas:whenthepowerexceeds1.5timesnormalpower,the speeddropsto50%oftheoriginalspeed.Additionally,theadaptivecontrolwithan interval of 13 r/min (AC(int=13)) is also added for comparison to show the resemblance ofrule-basedandadaptivecontrolwhentheintervalbecomesrelativelylarge.Therea-soning for choosing int=13 as a comparison is:in the current settings, since the available speed range is [6,25], when int=13, the next chromosome AC can select has a speed range of[6,19].Itscenterspeedis12.5r/min,whichisexactly50%oftherequestedspeed 25r/min.ThecomparisonresultsareindicatedinFigure4.14.Theexperimentresults suggestthat:firstofall,theoverallefficiencyof*int*=8ismuchlowerthenthatof *int* = 0.25;secondly,*int* = 8lacksofbasicflexibilitysinceitwillnotresumetooriginal requested speed when the torque is withdraw.This is not to be preferred in the industry, sincealotofefficiencyislostafterthisspeedcontrol.

TheresultsofthespeedperformanceinFigure4.14aindicate,firstofall,interms ofthereactiontimewhenreactingtothetorquechanges,wehave:AC(int=0.25)< rulebased<AC(int=13).Thankstothesmallinterval,thePIDerrorcalculationin AC(int=0.25)hasaverysmallthresholdsothatitcanactreactfastesttotorquein-crease at 4.323s.Rule based mechanism reacts later at 11.678s as soon as the pre-defined 1.5timespowerthresholdisreached.Notethattherulebasedmechanism’sreaction speedpurelydependsonthepowerthreshold,asmallerthresholdwillleadtoshorter reactiontimehoweverlesstoleranceforthepowerincrease.TheAC(int=0.25)react theslowestat17.770s,sincethePIDerrorcalculationhasaverylargethreshold.Sec-ondly,rulebasedandAC(int=13)exhibitbadflexibilityintermsofspeed,thespeed willnotbouncebackwhenthetorqueiswithdrawn.Forrule-based,thisisnormalsince rule-basedmechanismsareusuallydesignedforfail-safetoprotectthemachine. As forAC(int=13),itsinflexibilityisalsoduetothelargePIDerrorthreshold,thesame reasonasinlongreactiontime.Thirdly,thehighestspeedtheAC(int=0.25)canreach aftertheabnormaltorqueiswithdrawnis20.375r/min,whichislowerthanthesame experimentonano-noisesimulationmotor.Theexistenceofnoises(friction,current sensorreadinginaccuracy,etc.)hasslightlydegradedtheperformanceofadaptivecon-trol since the PID error calculation depends on the real-time power and velocity readings.

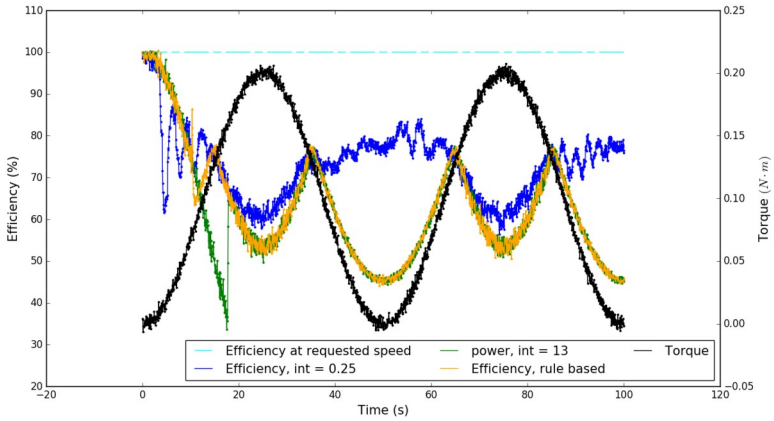
Furthermore,fromtheresultsofthepowerperformanceinFigure4.14b,wecansee that:inthefirstplace,allthepowercurvesarefluctuatingaroundthepowerconsump-



(a)The comparison of speed among int=0.25, int=13 and rule based systems



(b)Thecomparisonofpoweramongint=0.25,int=13andrule basedsystems



(c)Thecomparisonofefficiencyamongint=0.25,int=13andrule basedsystems

Figure4.14:Performancecomparisonsamongint=0.25,int=13andrulebasedsystems

Items	AC(int=0.25)	Rulebased	AC(int=13)
Efficiency(%)	73.2	63.0	61.8

Table4.8:Comparisonsofaverageefficiencyamongint=0.25,int=13andrulebased systems

tion at requested speed, giving roughly similar average power results.However rule based andAC(int=13)hasamuchlargervariancethanAC(int=0.25),sinceaftercontrol,the speedremaineduncharged,thusthepowerconsumptionchangeswiththetorqueina positivecorrelationmanner.Whereasouralgorithmmanagestomitigatetheinfluence oftheappliedabnormaltorquebychangingthespeedwiththetorqueinanegative correlationmanner.Inthesecondplace,forrulebasedandAC(int=13),thegraphalso showanobviouspowersurgebeforethethresholdsarereached,whichisnotgoodfor thesystem.WhileAC(int=0.25)managetoavoidthisthankstoitssmallthreshold.

Lastbutnotleast,theefficiencyresultsinFigure4.14candTable4.8provideacom-prehensiveviewoftheperformanceofthreedifferentsystems:firstofall,AC(int=0.25) hasthebestaverageefficiency.Therankingforaverageefficiencyis:AC(int=0.25)> rulebased>AC(int=13).Secondly,AC(int=0.25)hasbetterefficiencyperformance when the abnormal torque is just applied.Before the speed begins to decrease, all of the algorithms exhibit significant efficiency drop caused by the power consumption increase. Themagnitudesofthedroplargelydependonthereactiontime,thereforehasthefol-lowingranking:AC(int=0.25)≤rulebased<AC(int=13).Thirdly,AC(int=0.25)also hasbetterperformancewhentheappliedabnormaltorquereachesitspeak.Thisisbe-cause,forrulebased,thetargetspeedaftercontrolisfixed(10r/min);forAC(int=13), duetoitslargeinterval,thespeedchoiceisrestricted,thusalsofixed(10r/min).While forAC(int=0.25),thespeedcanbeevensmallerwhenthespeedreaches10r/minif thetorquekeepsincreasing.Lastbutnotleast,AC(int=0.25)hasthebestperformance when the torque is withdrawn.In rule-based, this functionality is not implemented since itonlyservesasafail-safe;inAC(int=13),thisisalsonotpossiblebecauseofitsspeed choicerestriction.WhileinAC(int=0.25),thespeedcanreturnto20.125r/min,which istheclosesttotheoriginallyrequestedspeed,thusincreasetheefficiencycomparedto rulebasedandAC(int=13).

Ingeneral,wecanconcludethatourimplementedadaptivecontrolalgorithmusing anintervalof0.25hassignificantadvantagesovertherulebasedmechanisminterms ofspeed,powerandefficiency.TheperformanceofAC(int=13)alsoshowsthat,ifwe increasetheintervaltobeverylarge,itwillhavesimilarbehavioursastherulebased systemhoweverwithmuchlongerreactiontime.

5Conclusion

Thischaptersummarizesthewholemasterthesiswork.Firstofall,theconceptsand ideasbehindthetopicareintroduced,thenthemethodologyofimplementingthealgorithmsarerecapitulated.Afterward,theexperimentsandevaluationsarefurtheranalyzed.Finally,possibledirectionsforfutureworkandpotentialimprovementmethods areproposed.

5.1Summary

Inthisthesis,aconveyorbelttestbenchwithmultiplesensorsisimplementedanda real-time predictive maintenance mechanism is developed to detect whether the conveyor beltisfunctioningnormallyorthereisanabnormalbehaviorthatneedsattention.Relatedworkssuggestthatanomalydetectionapproachescanbeusedforsuchdetection. Despitethefactthattheconceptsofpredictivemaintenanceandanomalydetectionare rarelycombined,itispossibletomodifyanomalytechniquestosuitourrequirements.

Wefirstcollectdatafromtherealservomotoronourconveyorsystem.Thesedata arenotonlyusedtotrainthepredictivemaintenancemodelsbutalsousedtocreate a“digitaltwin”oftherealmotor,i.e.,asimulatedmotormodelthathassimilarcharacteristicsastherealmotor.Thissimulationmodelislaterusedinbothpredictive maintenanceandadaptivecontrolfortestsandexperimentstoavoidunwantednoises that can be caused by, for example, imbalanced friction on the conveyor belt, glitches on the motor gears or any interior part, the inaccuracy of the current sensor.Then, the im-plementation of predictive maintenance algorithm is carried out.Three popular anomaly detectionalgorithms,namelyone-classSVM,*k*-NNandkernelPCA,areimplemented inasemi-supervisedmanner:first,themodelthatcapturesthecharacteristics(normal speed-powerrange)ofthemotoristrained;then,usingthetrainedmodel,areal-time anomaly detector is implemented.Later on, various experiments are conducted to select thebestparametersforeachalgorithmandevaluatetheirperformance.Toovercome theimbalancedclassdistributionprobleminparametertuninginouralgorithms,aself-adaptivedatashiftingmethodfrom[2]isusedtogeneratepseudotargetdataaswellas artificialoutlierdata.Comparingthreealgorithmswiththebestparameters,one-class SVM,*k*-NNandkernelPCAachieveaclassificationaccuracyof99.63%,49.93%and 97.46%,respectively.Regardingtrainingtime,theytake0.17s,0.0034sand65.281s, respectively,underthesameconditions.Andforthepredictingtime,theyeachspend 0.062s,0.538sand1.758s.Resultsshowthatone-classSVMachievethebestclassificationaccuracy,whileusingrelativelylesstimeinbothtrainingandpredicting.Selected

as the best algorithm, one-class SVM is compared with the rule-based mechanism in predictivemaintenancespeed.Theexperimentresultsshowthatwhenspeed=25r/min, one-classSVMhasa69.17%fasterresponsethantherule-basedsystem.

Although the proposed predictive maintenance algorithm can effectively detect the faults inthesystemorthefluctuationsofloadtorque,itcannotdirectlycontrolthesystem toreactaccordingly.Therefore,weimplementanovelregressionbasedadaptivecon-trolmechanismthattakesdecisionstoadjusttheoperationmodeinawaythatkeeps theoperationrunningandprotectsthemotorandtheproductionlinefromapossible break-down.Relatedworksforadaptivecontrolalgorithmsmainlyfocusonimprov-ingspeed/ positioncontrolperformance(e.g.,PIDcontrollerparametertuningornovel controllerdesign)underparametricuncertaintyconditions(e.g.,extranoiseorvarying operatingenvironment). Thesealgorithms,especiallymeta-heuristicalgorithmssuch asgeneticalgorithm,achievedgoodperformanceinthestate-of-the-art.Therefore,we notedthepotentialofmigratingsuchsolutionstoourusecase.

Inthisdirection,ageneticalgorithmbasedreal-timeadaptivecontrolisproposedand implemented.Inthefirstplace,normalrunningdatacollectedfromthemotorareused tobuildapolynomialregressionmodelforthecalculationofthepower. Then,the systemcalculatesthePIDerroranddecideswhetherthegeneticalgorithmshouldbe triggeredisimplemented.IftheGAiscalled,itselectsanewspeedvaluefortheopera-tionasareactiontothedrasticallyorunexpectedlychangingloadtorque.Weconduct experimentstodecideonthebestparametercombinations.Intheend,thealgorithm with the best parameters is compared with the state-of-art rule based algorithms to show the effectiveness of our approach against preventive or rule-based soluti ons that usually terminatestheoperationorrespondstoolatebeforeabreak-down.Theresultsshowed thatouralgorithmexhibitedmuchhigherspeedflexibility,lowerpowerdevianceand 16%percenthigherefficiency.

5.2FutureWork

The results of the experiments clearly show that our algorithm significantly outperforms therule-basedmechanisms,andithasthepotentialtoperformwellinourexperiment setups.Therefore, the primary future work is to apply the current algorithm to the servo motorinourexperimentalsetup.Moreover,therearestillsomeunsolvedissuesthat requireadditionalattention.Forexample,intheadaptivecontrolalgorithm,whenthe torque is increasing, the speed decreasing process exhibits small fluctuations.Therefore, itissuggestedtotakeadditionalmeasurestosmoothenthespeedbehaviours.Lastbut notleast,acompletemechanismintegratingbothpredictivemaintenanceandadaptive controlneedstobeproposedtobetterdemonstratethefunctionalityandeffectiveness ofthesystem.

ListofAcronyms

AC AdaptiveControl
AI ArtificialIntelligence
I4.0 Industry4.0
EPD EdgePatternDetection
IoT InternetofThings
k-NN k-nearestNeighbor
PCA PrincipleComponentAnalysis REST
RepresentationalStateTransfer ROS
RobotOperatingSystem
SVM SupportVectorMachine

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